

SYSTEM REQUIREMENTS SPECIFICATION

DIGITAL INTERLOCKING (DIL)

Digital Interlocking OBB

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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APPROVAL		
	Name and Role/Job Title	E-Signature
Written by	Eber G. / RM	
Verified by	Janneschitz C. / QAM	
Approved by	Eichberger M. / PDA	

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LOG OF CHANGES			
Edition	Date	Author	Change Description
01	2025-05-xx	Hitachi Rail GTS Austria / EJT RM G. Eber	First edition. Details see 4.1

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CONTENTS

1.	INTRODUCTION	6
1.1	Purpose	6
1.2	Intended Audience	6
1.3	Distribution	6
1.4	Document Management	6
1.5	Referenced Documents	6
1.6	Abbreviations and Acronyms	7
1.7	Structure of the Requirements	7
1.8	Information for Document Generation	7
1.9	Open Points	8
2.	OVERVIEW	9
3.	REQUIREMENTS.....	11
3.1	D2.3 General Requirements	11
3.1.1	Architecture.....	11
3.1.2	Automatic Self Test.....	15
3.1.3	Environment and Protection	18
3.1.4	EULYNX Implementation	46
3.2	D2.5 ELEKTRA 4.0 (DIL-CU).....	50
3.2.1	Architecture.....	50
3.2.2	Interfaces to Adjacent Systems	63
3.2.3	System Startup	73
3.3	D2.2 RAMS.....	82
3.3.1	General.....	82
3.3.2	Reliability	85
3.3.3	Maintainability and Maintenance	93
3.3.4	Availability	107
3.3.5	Safety.....	131
3.3.6	Miscellaneous	140
4.	HISTORY.....	146
4.1	Edition 01.....	146
4.1.1	Proposal 01.....	146
APPENDIX A	DECISIONS.....	147
APPENDIX B	INDEX OF REQUIREMENTS	148

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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LIST OF TABLES

Table 1: Reference Documents 7

Table 2: Abbreviations and Acronyms..... 7

Table 3: Document Generation Settings 8

Table 4: DIL: Safety Requirements..... 132

Table 5: SIL-Allocation of safety-related Part Systems and Interfaces..... 132

LIST OF FIGURES

Figure 1: System Overview 10

Figure 2: DI-CU: Logical System Border 51

Figure 3: Software Change 54

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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1. INTRODUCTION

1.1 PURPOSE

The present document contains the system specification of the ELEKTRA 4.0 system Digital Interlocking (DIL). It is developed in the context of the “Generic Application Digital Interlocking OBB” (DIL).

1.2 INTENDED AUDIENCE

- Product Design Authority
- Technical Project Managers
- Solution Engineering Managers
- Product Line Manager
- Test Manager
- Validator
- Verifier
- Safety Assurance Manager
- Quality Assurance Manager
- RAM Assurance Manager
- System Architect

1.3 DISTRIBUTION

A new version of the document is distributed to all persons according to chapter 1.2

1.4 DOCUMENT MANAGEMENT

See Requirements Management Plan [DIL_RMP] for a description of the SSS document management.

1.5 REFERENCED DOCUMENTS

TO COMPLETE.

[SP_GLOSS]	SharePoint Project Glossary https://urbanandmainlines.sharepoint.com/sites/AT-ELEKTRA4.0/Lists/Glossary%20NEW/
[EL_GLOSS]	Glossary: Definitions, Acronyms and Abbreviations - ELEKTRA 2.x System, 3BU 15000 0081 DCAPA, Ed. 06
[DIL_RMP]	Requirements Management Plan – Digital Interlocking OBB, 3BU 17300 1207 DUAPA, Ed. 01
[LH_PART_D]	DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte, 3BU 17300 0004 DSAPC, Ed. 03

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[PH_PART_D2.2]	Pflichtenheft Digital Interlocking OBB - D2.2 Teilpflichtenheft RAMS, 3BU 17300 0022 DTAPC, Ed. 01
[PH_PART_D2.3]	Pflichtenheft Digital Interlocking OBB - D2.3 Teilpflichtenheft Allgemeine Anforderungen, 3BU 17300 0023 DTAPC, Ed. 01
[PH_PART_D2.5]	Pflichtenheft Digital Interlocking OBB - D2.5 Teilpflichtenheft Teilsystem DSTW-ZE, 3BU 17300 0025 DTAPC, Ed. 01
...	...

Table 1: Reference Documents

1.6 ABBREVIATIONS AND ACRONYMS

See Definitions, Acronyms and Abbreviations [SP_GLOSS] and [EL_GLOSS].

Additional abbreviations not mentioned there: **see table below.**

...	...
-----	-----

Table 2: Abbreviations and Acronyms

1.7 STRUCTURE OF THE REQUIREMENTS

See [DIL_RMP] for a description of the foreseen SSS requirement structure.

1.8 INFORMATION FOR DOCUMENT GENERATION

The present document was generated by using IBM™ Rational DOORS.

Settings applied:

DOORS/RMF Version:	DOORS 9.7.2.5/RMF 6.1.0.9
DOORS/RMF Module Name:	EL4 SSS
DOORS/RMF Module Version (Baseline):	1.0
DOORS/RMF Module Path:	/ELEKTRA 4.0/Specification
DOORS/RMF Export View:	WEXP SSS 1251 (DIL)
DOORS/RMF Current Filter:	(TA Req Status != Obsolete) AND (TA SRS includes 1251)
DOORS/RMF Export Options 1:	2,false,0,0,false,false,0,true,false,
DOORS/RMF Export Options 2:	true,false,true,2,false,false,true,PostProcessAllStyles,\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_1251.log,
DOORS/RMF Export Options 3:	\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-

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	TEMPLATES\EL4\SRS\Exports\WEXP_1251.log,t rue,true,true,false,false,true,0,true,4,
DOORS/RMF Export Options 4:	0,###,===,true,false,false,true,true,
Documentation-Vob Template Name:	3BU_17300_1251_DTAPA-Template

Table 3: Document Generation Settings

1.9 OPEN POINTS

TODO.

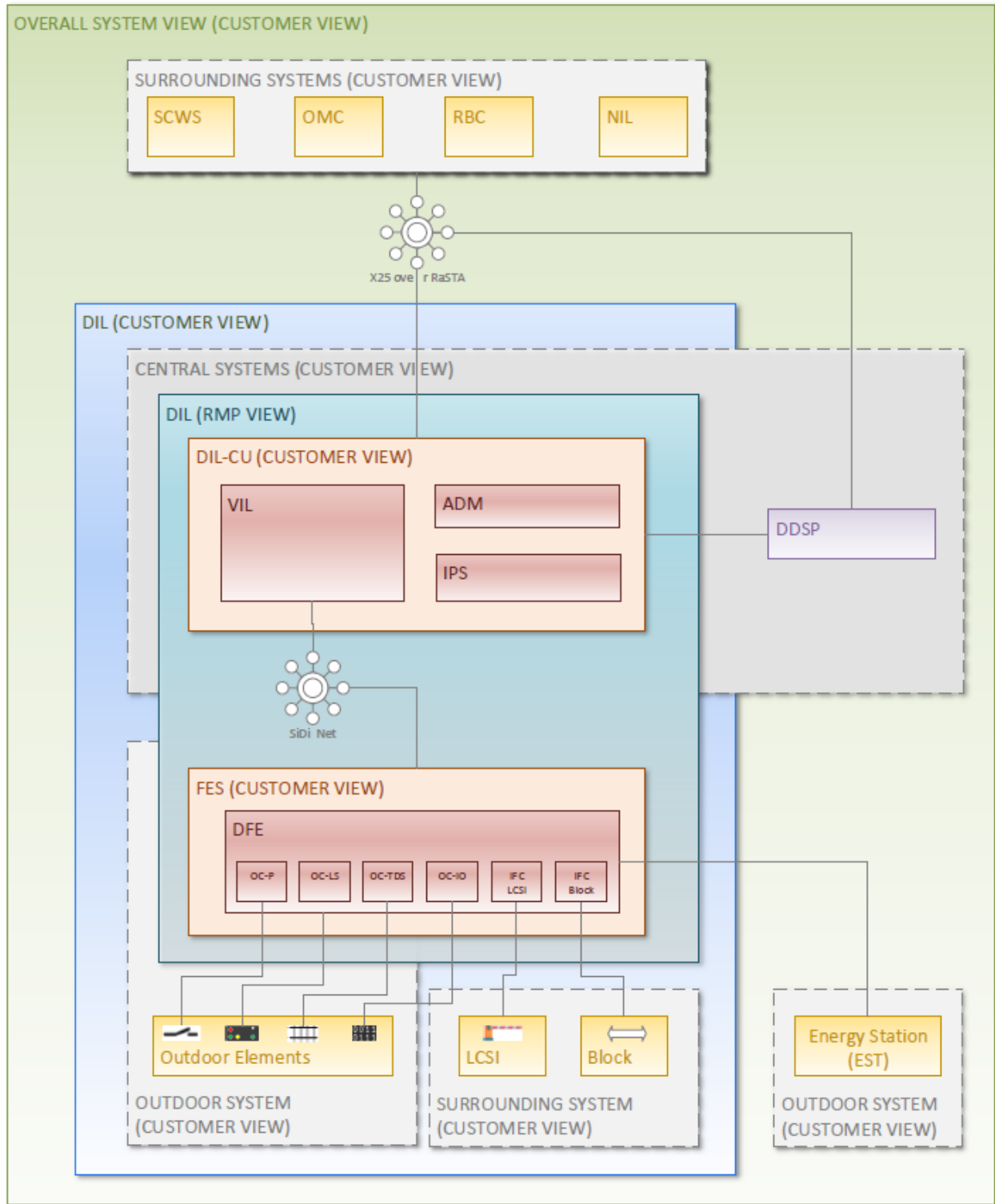
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2. OVERVIEW

The following picture gives an overview of the overall system and the [DIL] system considering both the technical project view and the customer's view.



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Figure 1: System Overview

This picture will be moved later to the [DIL_RMP].

TO COMPLETE: Short overview of DIL system.

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3. REQUIREMENTS

3.1 D2.3 GENERAL REQUIREMENTS

3.1.1 ARCHITECTURE

3.1.1.1 OVERVIEW

Refer here also to chapter [2] of this document.

The **Digital Interlocking System** [DIL] consists of the the following 3 main components:

- Central Systems
- Network
- Outdoor System

The **Central Systems** and the **Surrounding Systems** include the following systems:

- Data centre with [DIL-CU] and the peripheral systems [DDSP] and [SSP]
- Signal-controlled warning system ([SCWS]/[SCWS-S])
- Operations control centres ([OMC])
- Radio Block Center ([RBC])
- Neighbour [IL]s
- Level crossing safety installation ([LCSI])

== **MISMATCH** ==

In contrast to the document [D2.3 Teillastenheft Allgemeine Anforderungen] but according to the document [D1 Leistungsbeschreibung] the [LCSI] is also considered as "Surrounding System" and not as "Outdoor System".

The **Outdoor System** includes the following systems:

- Field element stations ([FES]) with object controllers ([OC]s) for the point, light signal, track vacancy detection, generic input and output subsystems
- Energy station ([EST])

== **MISMATCH** ==

In contrast to the document [D2.3 Teillastenheft Allgemeine Anforderungen] but according to the document [D1 Leistungsbeschreibung] the [LCSI] is also considered as "Surrounding System" and not as "Outdoor System".

3.1.1.2 CENTRAL SYSTEMS

The interlocking logic is implemented in **Virtual Interlockings** [VIL]s, which form the [DIL-CU] with server hardware located in the Data Centre.

The **Data Diagnostic Service Platform** [DDSP] and **Security Services Platform** [SSP] systems are peripheral systems located in the data centre that are necessary for the functionality of the [DIL]. Their functionality is described in separate specifications based on the [EULYNX] specifications.

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3.1.1.3 OBJECT CONTROLLER

Object Controller [OC] exist for the following **Outdoor Elements**:

- Point [P]
- Light Signal [LS]
- Train Detection System [TDS]
- Generic I/O [GenIO]

EL4-DIL-SSS-REQ-00819 OC - Reserve Computing Power (Draft)	
The [DIL] shall provide sufficient reserve computing power for [OC] enhancements (update/upgrade).	
Comments: == MISMATCH == Das ist eine Anforderung, aber im LH nur als "Info" markiert.	
Category: GCR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation:	Priority: Mandatory Origin: ==> [EL4 PH] [03_101] 2.4.0-2 Für Erweiterungen des OC (Update/Upgrade) muss genügend Rechenleistung als Reserve vorhanden... Scope: System Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00737 OC - Status Messages - 1 (Draft)	
The [OC] for [LS], [P] and [Gen-IO] shall support the following status messages: •Normal operation •Data traffic via ETH interface •Maintenance/fault mode •Boot process •Initialisation module	
Comments: <u>Note:</u> Further details will be defined by the supplier with the customer during the specifications phase.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Priority: Mandatory Origin: ==> [EL4 PH] [03_100] 2.4.0-1 Folgende Zustände müssen mindestens am Objekt Controller selbst erkennbar sein und eindeutig... Scope: System Verification: System test See also:
Issues/Decisions:	

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EL4-DIL-SSS-REQ-00807		OC - Status Messages - 2		(Draft)			
In addition, the [OC] for [LS], [P] and [Gen-IO] shall indicate the status of the interface to the external elements:							
•Input ports: •Input signal OK / no input signal •Error at the corresponding port •Output ports: •Output switched on and OK / not switched on •Error at the corresponding port							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_100] 2.4.0-1 Folgende Zustände müssen mindestens am Objekt Controller selbst erkennbar sein und eindeutig...	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00		Verification:		System test	
		DIGITAL INTERLOCKING (DIL)		See also:			
		DIGITAL FIELD ELEMENTS (DFE)					
Issues/Decisions:							

EL4-DIL-SSS-REQ-00808		OC - Status Messages - 3		(Draft)			
In addition, the [OC] for [P] shall support the following status messages:							
•Point in left / right end position •Left / right end position in simulation mode •Point movement •Circulation in simulation mode							
•Point forced							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_100] 2.4.0-1 Folgende Zustände müssen mindestens am Objekt Controller selbst erkennbar sein und eindeutig...	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)		Verification:		System test	
				See also:			
Issues/Decisions:							

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EL4-DIL-SSS-REQ-00809		OC - Status Messages - 4		(Draft)
The [OC] for [TDS] shall support the following status messages: •Normal operation •Data traffic via ETH interface •Status of the connection between the central computers •Malfunction (the type of malfunction is indicated by a blink code)				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_100] 2.4.0-1 Folgende Zustände müssen mindestens am Objekt Controller selbst erkennbar sein und eindeutig...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00806		OC - Status Messages - 5		(Draft)	
In addition, the [OC] for [TDS] shall indicate the status of the interface to the outdoor system:					
•Input / output card is OK / not OK •Connection to outdoor element OK / not OK •Data transfer to/from outdoor element					
Comments:					
Category: FUR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [03_100] 2.4.0-1 Folgende Zustände müssen mindestens am Objekt Controller selbst erkennbar sein und eindeutig...			
Target Release: PV_EL_GEN_2.20.0		Scope: System			
BL_OBB_40.00					
Allocation: DIGITAL INTERLOCKING (DIL)		Verification: System test			
DIGITAL FIELD ELEMENTS (DFE)		See also:			
Issues/Decisions:					

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EL4-DIL-SSS-REQ-00738		OC - Workload		(Draft)	
The [CPU] workload (computing power) of the [OC] shall not exceed 60% during normal operation when it is first integrated to the customer's network.					
Comments:		<u>Note:</u> The [CPU] will not reach the required maximum workload when it is first admitted to the customer's network. In a further phase, more powerful [CPU]s with a maximum [CPU] workload of 60% will be integrated into the object controllers.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_102] 2.4.0-3 Die CPU-Auslastung (Rechenleistung) des OC darf bei Erstzulassung im Netz des AG im Regelbet...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test		
Issues/Decisions:		See also:			

EL4-DIL-SSS-REQ-00739		OC - Reserve Capacity		(In negotiation)	
The supplier shall inform the customer if enhancements to the [OC] (update/upgrade) exceed the reserve computing capacity.					
Comments: == OPEN POINT == This is a process requirement.					
Category:	MCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_103] 2.4.0-4 Der AN muss den AG informiern falls Erweiterungen des OC (Update/Upgrade) die Reserve der Re...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
	BL_OBB_40.00				
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test		
		See also:			
Issues/Decisions:					

3.1.2 AUTOMATIC SELF TEST

The self-tests serve to carry out the recurring inspections to determine the proper condition of the subsystems and of the [DIL].

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EL4-DIL-SSS-REQ-00740		Automatic Self Test - System Level		(Draft)			
The entire [DIL] shall be able to perform an automatic self-test.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_291] 2.5.0-1.0-1 Das Gesamtsystem DSTW muss einen automatischen Selbsttest durchführen können.	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)		Verification:		System test	
				See also:			
Issues/Decisions:							

EL4-DIL-SSS-REQ-00741		Automatic Self Test - Subsystem Level		(Draft)	
Each subsystem of the entire [DIL] shall be able to perform an automatic self-test.					
Comments:					
Category:	FUR	Priority:	Mandatory	==> [EL4 PH] [03_285] 2.5.0-1 Jedes Teilsystem des Gesamtsystems DSTW muss einen automatischen Selbsttest durchführen könn...	
Originator:	Customer	Origin:			
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification: See also:	System test		
Issues/Decisions:					

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EL4-DIL-SSS-REQ-00742		Automatic Self Test - Initiation by DDSP		(Draft)	
It shall be possible to initiate the self-tests of the subsystems of the [DIL] and of the [DIL] via the [DDSP].					
Comments:					
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_287] 2.5.0-2.0-1 Die Selbsttests der Teilsysteme des Gesamtsystems DSTW sowie des Gesamtsystems DSTW müss...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test		
		See also:			
Issues/Decisions:					

EL4-DIL-SSS-REQ-00743		Automatic Self Test - Reporting of Result		(Draft)	
The result of the self-test shall be a standardised report on the condition of the subsystems and the [DIL].					
Comments:		The exact contents of the report will be agreed between the supplier and the customer in the specifications phase.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_288] 2.5.0-2.0-2 Das Ergebnis des Selbsttests muss ein standardisierter Report über den Zustand der Teils... ==> [EL4 PH] [03_289] 2.5.0-2.0-3 Die genauen Inhalte des Reports sind zwischen AN und AG in der Pflichtenheftphase abzust...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test		
		See also:			
Issues/Decisions:					

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EL4-DIL-SSS-REQ-00744		Automatic Self Test - Repeatability		(Draft)			
It shall be possible to repeat the self-test for the subsystems and for the [DIL].							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_290] 2.5.0-2.0-3.0-1 Die wiederholte Ausführung des Selbsttest für die Teilsysteme des Gesamtsystems DSTW...	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)		Verification:		System test	
				See also:			
Issues/Decisions:							

3.1.3 ENVIRONMENT AND PROTECTION

3.1.3.1 ENVIRONMENTAL CONDITIONS

3.1.3.1.1 GENERAL

From customer side there are no special requirements for the outdoor air volume flow rate for the housings of the [FES] and the field elements.

== **MISMATCH** ==

This is considered as requirement in the LH

EL4-DIL-SSS-REQ-00594		Environmental Conditions - OEVE EN 50125-3		(Draft)			
All [DIL] components shall operate without restrictions according to the environmental conditions specified in [OEVE] [EN 50125-3].							
Comments:							
Category:		PER		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_129] 3.1.0-1 Die DSTW-Streckenausrüstung (Außenanlage) und die DSTW-ZE muss unter den in OEVE/OENORM EN 5...	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00		Verification:		System test	
		DIGITAL INTERLOCKING (DIL)		See also:			
		INTEGRATED PLATFORM					
		SIGNALLING (IPS)					
		DIGITAL FIELD ELEMENTS (DFE)					
		FES					
Issues/Decisions:							

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00595		Environmental Conditions - FES, OC - Height Class A1		(Draft)
All [FES] and [OC] components shall operate without restrictions in the height class A1 (until 1400 m) according to [EN 50125-3].				
Comments:				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_130] 3.1.0-2 Es muss die Höhenklasse A1 gemäß EN 50125-3 - bis 1400 m - berücksichtigt werden.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	DIGITAL FIELD ELEMENTS (DFE)			
	OC-P			
	OC-IO			
	OC-LS			
	OC-TDS			
	FES			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00596		Environmental Conditions - DIL - Climate Class T2 (Indoor)		(Draft)
All [DIL] components shall operate without restrictions in the climate class T2 according to [EN 50125-3] with the following values: 1. outdoors: -40 °C to +45 °C, 2. in the control cabinet: -25 °C to +65 °C, 3. in the container without temperature monitoring: -20 °C to +50 °C, 4. in the container with temperature monitoring: +15 °C to +30 °C, 5. in the building without air conditioning: -5 °C to +40 °C, 6. in the building with air conditioning: +18 °C to +27 °C.				
Comments:				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_131] 3.1.0-3 Je nach Standort der DSTW Komponenten müssen Temperaturen der Klimaklasse T2 gemäß EN 50125...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES	Verification:	System test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00597 Environmental Conditions - DIL - Climate Class T2 (Track Area) (Draft)			
In addition all [DIL] components <u>in the track area</u> shall operate without restrictions outdoors: -40 °C to +70 °C (including solar radiation).			
<i>Comments:</i>			
<i>Category:</i>	PER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_131] 3.1.0-3 Je nach Standort der DSTW Komponenten müssen Temperaturen der Klimaklasse T2 gemäß EN 50125...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00745 Environmental Conditions - FES - Temperature Change (Draft)			
The housings of the [FES] shall be designed in such a way that a temperature change in accordance with [EN 5012-3] Section 4.3 has no effect on the function of the [FES] components.			
<i>Comments:</i>			
<i>Category:</i>	PER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_132] 3.1.0-4 Bei Einwirken von Fahrtwind muss bei der Auslegung des Gehäuses die maximale Temperaturänder...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00746 Environmental Conditions - FES - Climate Class T2 (Draft)			
The housings of the [FES] shall be designed in such a way that the humidity of climate class T2 according to [EN 50125-3] with the locations according to table 3 has no effect on the function of the [FES] components.			
<i>Comments:</i>			
<i>Category:</i>	PER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_133] 3.1.0-5 Es muss die Luftfeuchtigkeit der Klimaklasse T2 gemäß EN 50125-3 mit den Orten gemäß Tabelle...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00747		Environmental Conditions - FES-S, FES-M - Protection Class IP 65 - 1		(Draft)
The housings of the field element stations [FES]-S and [FES]-M shall be designed according to protection class [IP]65 according to [EN 60529] and shall take into account climate class T2 according to [EN 50125-3] with a rainfall of 6 mm/min.				
Comments:				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_134] 3.1.0-6 Es muss die Klimaklasse T2 gemäß 50125-3 mit einer Regenmenge von 6 mm/min berücksichtigt we...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
Issues/Decisions:		See also:		

EL4-DIL-SSS-REQ-00748		Environmental Conditions - FES-S, FES-M - Protection Class IP 65 - 2		(In negotiation)
The housings of the field element stations [FES]-S and [FES]-M and the field elements shall be designed according to protection class [IP]65 according to [EN 60529] and shall protect the components inside from moisture penetration when wind and air streams occur in accordance with [EN 50125-3] section 4.7.				
Comments: == OPEN POINT == Consideration of field elements.				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_135] 3.1.0-7 Die Einhausungen der FES und der Feldelemente müssen die Komponenten im Inneren bei allen vo...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
Issues/Decisions:		See also:		

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00749		Environmental Conditions - FES-S, FES-M - Protection Class IP 65 - 3		(In negotiation)
The housings of the field element stations [FES]-S and [FES]-M and the field elements shall be designed according to protection class [IP]65 according to [EN 60529] and shall withstand hailstones with diameters of up to 15 mm in accordance with [EN 50125-3] section 4.7.				
Comments: == OPEN POINT == Consideration of field elements.				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_136] 3.1.0-8 Die Einhausungen der FES und der Feldelemente müssen Hagelkörnern mit Durchmessern von bis z...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00750		Environmental Conditions - FES-S, FES-M - Protection Class IP 65 - 4		(In negotiation)
The housings of the field element stations [FES]-S and [FES]-M and the field elements shall be designed according to protection class [IP]65 according to [EN 60529] and shall permanently protect the components inside against moisture ingress according to [EN 50125-3] clause 4.7 for all types of ice that may occur.				
Comments: == OPEN POINT == Consideration of field elements.				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_137] 3.1.0-9 Die Einhausungen der FES und der Feldelemente müssen die Komponenten im Inneren bei allen vo...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
See also:				
Issues/Decisions:				

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00751		Environmental Conditions - FES - Solar Radiation		(Draft)
The [FES] components and the field elements shall be suitable for solar radiation up to 1090 W/m² in accordance with [EN 60721-3-4] class 4K26 "outdoors" (table 1).				
Comments:				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_138] 3.1.0-10 Die Einhausungen der FES und der Feldelemente müssen für die Sonnenstrahlung gemäß der [EN ...	
			==> [EL4 PH] [03_139] 3.1.0-11 Laut EN 60721-3-4 ist die zu erwartende maximale Sonnenstrahlungsleistung 1090 W/m².	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
	BL_OBB_40.00			
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test	
	DIGITAL FIELD ELEMENTS (DFE)	See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00752		Environmental Conditions - FES-S, FES-M - UV Radiation		(In negotiation)
The housings of the field element stations [FES]-S and [FES]-M and the field elements shall be designed to be resistant to [UV] radiation for the operational service life of 30 years.				
Comments: == OPEN POINT == Consideration of field elements.				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_140] 3.1.0-12 Die Einhausungen der FES und der Feldelemente müssen für die betriebliche Nutzungsdauer von...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00753		Environmental Conditions - FES - Pollution Classes		(In negotiation)	
The housings of the [FES] and the field elements shall be designed in such a way that they take into account the pollution classes					
•4 C 2 for chemical substances, •4 B 1 for biological substances, and •4 S 2 for mechanically active substances according to [EN 50125-3], table 4.					
Comments: == OPEN POINT == Consideration of field elements.					
Category:	PER	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_142] 3.1.0-14 Gemäß EN 50125-3 - Tabelle 4 muss der Verschmutzungsgrad mittel berücksichtigt werden.		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test		
Issues/Decisions:		See also:			

EL4-DIL-SSS-REQ-00810		Environmental Conditions - FES-L, FES-XL		(Draft)	
The housings of the field element stations [FES]-L and [FES]-XL shall be designed according to [RW 11.01.02] to fulfill the requirements according to [EL4-DIL-SSS-REQ-00747], [EL4-DIL-SSS-REQ-00748], [EL4-DIL-SSS-REQ-00749], [EL4-DIL-SSS-REQ-00750] and [EL4-DIL-SSS-REQ-00752].					
Comments:					
Category: PER		Priority: Mandatory			
Originator: Customer		Origin:		==> [EL4 PH] [03_134] 3.1.0-6 Es muss die Klimaklasse T2 gemäß 50125-3 mit einer Regenmenge von 6 mm/min berücksichtigt we... ==> [EL4 PH] [03_135] 3.1.0-7 Die Einhausungen der FES und der Feldelemente müssen die Komponenten im Inneren bei allen vo... ==> [EL4 PH] [03_136] 3.1.0-8 Die Einhausungen der FES und der Feldelemente müssen Hagelkörnern mit Durchmessern von bis z... ==> [EL4 PH] [03_137] 3.1.0-9 Die Einhausungen der FES und der Feldelemente müssen die Komponenten im Inneren bei allen vo... ==> [EL4 PH] [03_140] 3.1.0-12 Die Einhausungen der FES und der Feldelemente müssen für die betriebliche Nutzungsdauer von...	
Target Release: PV_EL_GEN_2.20.0		Scope:		System	
BL_OBB_40.00					
Allocation: DIGITAL INTERLOCKING (DIL)		Verification:		System test	
DIGITAL FIELD ELEMENTS (DFE)		See also:			
Issues/Decisions:					

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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3.1.3.1.2 ENVIRONMENTAL CONDITIONS IN EXISTING OPERATING ROOMS

EL4-DIL-SSS-REQ-00754 Environmental Conditions - Existing Operatin Rooms (Draft)	
If the installation of the subsystems or system components of the [DIL] is planned in existing operating rooms, the supplier shall observe the environmental conditions of these rooms in addition to the listed environmental conditions and design the subsystems and system components according to these environmental conditions.	
<i>Comments:</i>	
<i>Category:</i> PER <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_144] 3.2.0-1 Ist der Einbau der Subsysteme oder der Systemkomponenten des DSTW in bestehende Betriebsräum...
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00755 Environmental Conditions - Necessary Documents (In negotiation)	
The supplier shall request the necessary documents and requirements/conditions from the customer on a project-specific basis.	
<i>Comments:</i> == OPEN POINT == This is a process requirement.	
<i>Category:</i> MCR <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_144] 3.2.0-1 Ist der Einbau der Subsysteme oder der Systemkomponenten des DSTW in bestehende Betriebsräum...
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.1.3.1.3 SPECIFIC ENVIRONMENTAL AND AMBIENT CONDITIONS

EL4-DIL-SSS-REQ-00756 Environmental Conditions - Outdoor Elements (Draft)	
The components of the outdoor system shall also work properly and be fully functional in specific environmental and ambient conditions.	
<i>Comments:</i>	
<i>Category:</i> PER <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_146] 3.3.0-1 Die Komponenten der Außenanlage müssen auch in spezifischen Umwelt- und Umgebungsbedingungen...
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00757 Environmental Conditions - Agreement with Customer (In negotiation)	
The specific environmental and ambient conditions shall be specified in detail by the supplier/manufacturer and agreed with the customer, unless specified by [EN 50125-3].	
<i>Comments:</i> == OPEN POINT == This is a process requirement.	
<i>Category:</i> MCR <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_146] 3.3.0-1 Die Komponenten der Außenanlage müssen auch in spezifischen Umwelt- und Umgebungsbedingungen...
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

3.1.3.1.4 EXCEEDING OR FALLING BELOW ENVIRONMENTAL LIMIT VALUES

EL4-DIL-SSS-REQ-00758 Environmental Conditions - Monitoring of Warning and Limit Values (Draft)	
The [DIL] shall monitor the environmental conditions with regard to warning and limit values.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_148] 3.4.0-1 Umweltbedingungen des DSTW müssen hinsichtlich der Warn- und Grenzwerte überwacht werden. Be...
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00759		Environmental Conditions - Monitoring - Warning Message		(Draft)
If a <u>warning value</u> is exceeded, the [DIL] shall generate a warning message to the [DDSP] operator and maintenance stuff ([DDSP] or [SAM]).				
Comments:		The exceeding of a warning value (= threshold value 1) corresponds to a condition that can lead - in the long term - to a safety shutdown or failure of the [DIL] (e.g. in the event of excessive temperature due to a fan failure). This condition does not directly affect operation.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_148] 3.4.0-1 Umweltbedingungen des DSTW müssen hinsichtlich der Warn- und Grenzwerte überwacht werden. Be... ==> [EL4 PH] [03_149] 3.4.0-2 Maßnahmen können sein: 1. eine Warnmeldung (Schwellenwert 1) an DDSP-Bediener und Instandhal... ==> [EL4 PH] [03_150] 3.4.0-3 Schwellenwert 1 entspricht einem Zustand, der langfristig zu einer Sicherheitsabschaltung bz...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
Issues/Decisions:		See also:		

EL4-DIL-SSS-REQ-00811		Environmental Conditions - Monitoring - Automatic Shutdown		(Draft)
If a <u>limit value</u> is exceeded, the [DIL] shall generate a fault message to the operator ([HMI]) and maintenance stuff ([DDSP] or [SAM]), and perform an automatic shutdown.				
Comments:		The exceeding of a limit value (= threshold value 2) corresponds to a condition that immediately leads to a safety shutdown or failure of the [DIL]. This corresponds to an operational impairment.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_148] 3.4.0-1 Umweltbedingungen des DSTW müssen hinsichtlich der Warn- und Grenzwerte überwacht werden. Be... ==> [EL4 PH] [03_149] 3.4.0-2 Maßnahmen können sein: 1. eine Warnmeldung (Schwellenwert 1) an DDSP-Bediener und Instandhal... ==> [EL4 PH] [03_150] 3.4.0-3 Schwellenwert 1 entspricht einem Zustand, der langfristig zu einer Sicherheitsabschaltung bz...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
Issues/Decisions:		See also:		

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00760 Environmental Conditions - Data Preparation of Warning and Limit Values (Draft)			
All warning and limit values according to [EL4-DIL-SSS-REQ-00758] and [EL4-DIL-SSS-REQ-00811] shall be made available by data preparation.			
<i>Comments:</i>			
<i>Category:</i>	DAT	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_148] 3.4.0-1 Umweltbedingungen des DSTW müssen hinsichtlich der Warn- und Grenzwerte überwacht werden. Be...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

3.1.3.1.5 TRAIN-INDUCED AERODYNAMIC LOADS

EL4-DIL-SSS-REQ-00761 Environmental Conditions - FES - Train-induced aerodynamic Loads - Open Track - 1 (Draft)			
The housings of the [FES] and the field elements shall be designed with regard to the train-induced pressure and flow loads to which they are exposed on the open track.			
<i>Comments:</i>			
<i>Category:</i>	PER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_152] 3.5.0-1 Die Einhausungen der FES und der Feldelemente müssen hinsichtlich der zuginduzierten Druck- ...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00762 Environmental Conditions - FES - Train-induced aerodynamic Loads - Open Track - 2 (Draft)			
The strength of the housings of the [FES] and the field elements with respect to the train-induced pressure and flow loads on the open track shall be verified by calculation or empirically.			
<i>Comments:</i>			
<i>Category:</i>	TVR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_152] 3.5.0-1 Die Einhausungen der FES und der Feldelemente müssen hinsichtlich der zuginduzierten Druck- ...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

Hitachi Rail – Internal

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EL4-DIL-SSS-REQ-00763 Environmental Conditions - FES - Train-induced aerodynamic Loads - Tunnel - 1 (Draft)			
The housings of the [FES] and the field elements shall be designed with regard to the tension-induced pressure and flow loads to which they are exposed in the tunnel.			
<i>Comments:</i>			
<i>Category:</i>	PER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_154] 3.6.0-1 Die Einhausungen der FES und der Feldelemente müssen hinsichtlich der zuginduzierten Druck- ...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00764 Environmental Conditions - FES - Train-induced aerodynamic Loads - Tunnel - 2 (Draft)			
The strength of the housings of the [FES] and the field elements with regard to the tension-induced pressure and flow loads in the tunnel shall be verified by calculation or empirically.			
<i>Comments:</i>			
<i>Category:</i>	TVR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_154] 3.6.0-1 Die Einhausungen der FES und der Feldelemente müssen hinsichtlich der zuginduzierten Druck- ...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

3.1.3.1.6 WIND

EL4-DIL-SSS-REQ-00765 Environmental Conditions - Wind - FES - Housings (Draft)			
The housings of the [FES] and the field elements shall withstand the wind forces that occur.			
<i>Comments:</i>			
<i>Category:</i>	ECR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_156] 3.7.0-1 Die Einhausungen der FES und der Feldelemente müssen den auftretenden Windkräften standhalten...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00766		Environmental Conditions - Wind - FES - Claculation		(Draft)
The specifications in [EN 1991-1-4] shall be used to calculate the wind forces that occur.				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_156] 3.7.0-1 Die Einhausungen der FES und der Feldelemente müssen den auftretenden Windkräften standhalte...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00767		Environmental Conditions - Wind - FES - Determination		(Draft)
The determination of forces as the effect of aerodynamic loads on geometrically defined bodies shall be carried out in accordance with [EN 1991-1-4].				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_156] 3.7.0-1 Die Einhausungen der FES und der Feldelemente müssen den auftretenden Windkräften standhalte...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00768		Environmental Conditions - Wind - FES - Superposition		(Draft)
The housings of the [FES] and the field elements shall be capable of withstanding the superposition of aerodynamic loads and wind loads by selection of suitable combination coefficients.				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_157] 3.7.0-2 Die Einhausungen der FES und der Feldelemente müssen einer Überlagerung von zuginduzierten a...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
Issues/Decisions:		See also:		

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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3.1.3.2 ELECTROMAGNETIC COMPATIBILITY (EMC)

3.1.3.2.1 EMC TESTING

EL4-DIL-SSS-REQ-00598		EMC - Testing - System Level		(Draft)	
The [DIL] system, its subsystems and components shall undergo an [EMC] testing.					
Comments:		<u>Note:</u> Components provided by the customer shall be made available by the customer right in time for [EMC] tests of the corresponding subsystems. This requires eventually a requalification of subsystems under the responsibility of the customer.			
Category:		TVR	Priority:	Mandatory	
Originator:		Customer	Origin:	==> [EL4 PH] [03_163] 4.1.0-1 Das DSTW, seine Subsysteme und Komponenten müssen einer EMV-Prüfung unterzogen werden.	
Target Release:		PV_EL_GEN_2.20.0	Scope:	System	
Allocation:		BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES	Verification:	System test	
			See also:		
Issues/Decisions:					

EL4-DIL-SSS-REQ-00599		EMC - Testing - Subsystem/Component Level		(In negotiation)			
The [EMC] test for subsystems and components close to the track shall verify compliance with the [EMC] limits according to [EN 50121-4], whereby the assessment criterion A according to [EN 50121-1] shall be used for all limits.							
Comments: !! OPEN Der AG definiert den Begriff "DSTW" bestehend u.a. aus Komponenten der Innenanlage und Außenanlage. Im Dokument "D2.3 Teillastenheft Allgemeine Anforderungen", ID 03_250, wird jedoch eine Störfestigkeit nach EN50121-4 gefordert. Kann davon ausgegangen werden, dass dies für Komponenten im Rechenzentrum nicht erfüllt werden muss? !!							
Category: Originator: Target Release: Allocation:		TVR Customer PV_EL_GEN_2.20.0 BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES		Priority: Origin: Scope: Verification: See also:		Mandatory ==> [EL4 PH] [03_164] 4.1.0-2 Die EMV-Prüfung für Subsysteme und Komponenten in Gleisnähe muss die Einhaltung der EMV-Gren... System System test	
Issues/Decisions:							

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00600		EMC - Testing - Report		(Draft)			
The results of the [EMC] test shall be recorded in a test report together with a functional description of the system and shall be provided to the customer in an electronically modifiable form.							
Comments:							
Category:		DOR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_165] 4.1.0-3 Die Ergebnisse der EMV-Prüfung müssen zusammen mit einer Funktionsbeschreibung des Systems i...	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES		Verification:		System test	
				See also:			
Issues/Decisions:							

3.1.3.2.2 ELECTROMAGNETIC INTERFERENCE (EMI)

EL4-DIL-SSS-REQ-00601		EMC - EMI - EN 50121-4		(Draft)	
The [DIL] system, if necessary, its subsystems and components close to the track shall be compliant with the limits for [EMI] according to [EN 50121-4].					
Comments:					
Category:		PER	Priority:		Mandatory
Originator:		Customer	Origin:		==> [EL4 PH] [03_248] 4.2.0-1 Das DSTW, ggf. seine Subsysteme und Komponenten in Gleisnähe müssen die Grenzwerte für die S...
Target Release:		PV_EL_GEN_2.20.0	Scope:		System
Allocation:		BL_OBB_40.00	Verification:		System test
		DIGITAL INTERLOCKING (DIL)	See also:		
		INTEGRATED PLATFORM			
		SIGNALLING (IPS)			
		DIGITAL FIELD ELEMENTS (DFE)			
		FES			
Issues/Decisions:					

3.1.3.2.3 ELECTROMAGNETIC SUSCEPTIBILITY (EMS)

For use in the [DC], alternative verifications of [EMS] may be used.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00603		EMC - EMS - EN 50121-4		(Draft)
The [EMS] testing of the [DIL] system, its subsystems and components close to the track shall be performed according to the conditions listed in [EN 50121-4] chapter 6.2.				
Comments:				
Category:	TVR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_250] 4.3.0-1 Die Prüfung der Störfestigkeit des DSTW, seiner Subsysteme und Komponenten in Gleisnähe muss...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	DIGITAL FIELD ELEMENTS (DFE)			
	FES			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00604		EMC - EMS - DC Server		(Draft)
The [DC] server hardware shall meet at least the following requirements:				
•[Low Voltage Directive 2014/35/EU] (CE mark) - in the current version (including compliance with [EN 62368-1]) •[EMC Directive 2014/30/EU] •[Directive on hazardous substances 2011/65/EU] ("RoHS 2")				
•[Ecodesign Directive 2019/424/EU] (Lot 9)"				
Comments:				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_283] 4.3.0-2 Die Server-Hardware zum Einsatz im Rechenzentrum muss mindestens folgende Anforderungen erfü...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	DIGITAL FIELD ELEMENTS (DFE)			
	FES			
Issues/Decisions:				

3.1.3.2.4 RAIL CURRENTS

The current in the catenary is used as the basis for the magnitude of the rail currents.

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00769		EMC - Rail Currents - Continuous Current		(Draft)	
The functional and operational safety of the [FES], including the field elements, shall be ensured for rail currents with alternating current traction with a frequency of 16.7 Hz and 50 Hz for a continuous current <= 2 kAeff.					
Comments:					
Category:	SAR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_166] 4.4.0-2 Die Funktions- und Betriebssicherheit der FES inkl. der Feldelemente muss für Schienenströme...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test		
See also:					
Issues/Decisions:					

EL4-DIL-SSS-REQ-00770		EMC - Rail Currents - Short-Circuit Current		(Draft)	
The functional and operational safety of the [FES], including the field elements, shall be ensured for rail currents with alternating current traction with a frequency of 16.7 Hz and 50 Hz for a short-circuit current <= 40 kAeff for a duration of 100 ms.					
Comments:					
Category: SAR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [03_167] 4.4.0-3 Die Funktions- und Betriebssicherheit der FES inkl. der Feldelemente muss für Schienenströme...			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope: System			
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)		Verification: System test			
		See also:			
Issues/Decisions:					

3.1.3.2.5 EMC EMISSION OF VEHICLES

EL4-DIL-SSS-REQ-00771		EMC - Tele-Powering		(Draft)			
The [FES] housing, including the field elements, shall fully protect the components inside from high-frequency electromagnetic fields resulting from the tele-powering of Euro balises (Subset 036).							
Comments:							
Category:		ECR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_169] 4.5.0-1 Das Gehäuse der FES inkl. der Feldelemente muss die Komponenten im Inneren vor hochfrequente...	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)		Verification:		System test	
See also:				See also:			
Issues/Decisions:							

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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3.1.3.2.6 ELECTRIC STRENGTH OF CATENARY FOR TRACK COMPONENTS

EL4-DIL-SSS-REQ-00772		EMC - Catenary Voltage		(Draft)	
All track components shall be able to withstand the voltage of a catenary for placement near catenaries.					
Comments:					
Category:	ECR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_171] 4.6.0-1 Alle Gleiskomponenten müssen eine Oberleitungsspannungsfestigkeit für die Platzierung bei Ob...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test		
Issues/Decisions:		See also:			

3.1.3.3 GROUNDING AND SURGE PROTECTION

3.1.3.3.1 GROUNDING MEASURES

EL4-DIL-SSS-REQ-00605		Grounding and Surge Protection - Grounding Concept		(In negotiation)	
The supplier shall create, agree with the customer and implement an grounding concept in accordance with [EN 50310] and [EN 50122].					
Comments: == OPEN POINT == This is a process requirement.					
Category: MCR		Priority: Mandatory		Origin: ==> [EL4 PH] [03_173] 5.1.0-1 Der AN muss ein Erdungskonzept gemäß EN 50310 und EN 50122 erstellen, mit dem AG abstimmen u... Scope: System Verification: System test See also:	
Originator: Customer		Origin:			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:			
Allocation: DIGITAL INTERLOCKING (DIL)		Verification:			
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00606		Grounding and Surge Protection - DIL Integration		(Draft)	
The [DIL] system shall be integrated into the existing grounding system without causing any feedback.					
Comments:					
Category:	ECR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_174] 5.1.0-2 Das DSTW muss rückwirkungsfrei in die vorhandene Erdungsanlage eingebunden werden.		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES	Verification:	System test		
		See also:			
Issues/Decisions:					

3.1.3.3.2 LIGHTNING PROTECTION

EL4-DIL-SSS-REQ-00607		Grounding and Surge Protection - Lightning Protection Zone Concept		(In negotiation)	
The supplier shall create, coordinate with the customer and implement a lightning protection zone concept in accordance with [EN 62305].					
Comments: == OPEN POINT == This is a process requirement.					
Category: MCR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [03_176] 5.2.0-1 Der AN muss ein Blitzschutzkonzept gemäß EN 62305 erstellen, mit dem AG abstimmen und u...			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope: System			
Allocation: DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES		Verification: System test			
		See also:			
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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3.1.3.3.3 SURGE PROTECTION

EL4-DIL-SSS-REQ-00608		Grounding and Surge Protection - Surge Protection Concept		(In negotiation)	
The supplier shall create, coordinate with the customer and implement a surge protection concept in accordance with [EN 50124-2].					
Comments: == OPEN POINT == This is a process requirement.					
Category: MCR		Priority: Mandatory		Mandatory ==> [EL4 PH] [03_178] 5.3.0-1 Der AN muss ein Überspannungsschutzkonzept gemäß EN 50124-2 erstellen, mit dem AG abstimmen ... System System test	
Originator: Customer		Origin:			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:			
Allocation: DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE) FES		Verification: See also:			
Issues/Decisions:					

3.1.3.4 VIBRATIONS AND SHOCKS

3.1.3.4.1 PERMISSIBLE VIBRATIONS/SHOCKS – FIELD ELEMENT STATIONS

A suitable classification of mechanical conditions can normally be carried out in accordance with [EN 50125-3].

EL4-DIL-SSS-REQ-00773		Vibrations and Shocks - FES-S, FES-M - Function		(Draft)	
Vibrations and shocks shall not impair the function of a [FES]-S and [FES]-M regarding the limit values of the housing in which the [FES] under consideration is installed.					
Comments:					
Category:	ECR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_182] 6.1.0-1 Schwingungen und Stöße dürfen zu keiner Funktionsbeeinträchtigung einer Feldelementstation f...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
	DIGITAL FIELD ELEMENTS (DFE)				
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00774		Vibrations and Shocks - FES Components - Function		(Draft)
Vibrations and shocks shall not impair the function of any [FES] component (Object controller, the cable system and all elements of the outdoor system) regarding the limit values of the housing in which the [FES] components are installed.				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_184] 6.1.0-3 Schwingungen und Stöße dürfen zu keiner Funktionsbeeinträchtigung eines Teilsystems führen. ...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
Issues/Decisions:		See also:		

EL4-DIL-SSS-REQ-00775		Vibrations and Shocks - FES-S, FES-M - Impact Energy		(Draft)
The field element stations [FES]-S and [FES]-M shall be designed to resist an impact energy of 20 joules according to [IK]10 in accordance with [EN 62262].				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_186] 6.1.0-5 Die Feldelementstationen der Größe S und M müssen mit einer Stoßfestigkeit von IK10 gemäß EN...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
Issues/Decisions:		See also:		

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00814		Vibrations and Shocks - FES-L, FES-XL		(Draft)	
The field element stations [FES]-L and [FES]-XL shall be designed according to [RW 11.01.02] to fulfill the requirements according to [EL4-DIL-SSS-REQ-00773] and [EL4-DIL-SSS-REQ-00776].					
Comments:					
Category: ECR		Priority: Mandatory			
Originator: Technics (R&D)		Origin: ==> [EL4 PH] [03_182] 6.1.0-1			
		Schwingungen und Stöße dürfen zu keiner Funktionsbeeinträchtigung einer Feldelementstation f...			
		==> [EL4 PH] [03_186] 6.1.0-5 Die Feldelementstationen der Größe S und M müssen mit einer Stoßfestigkeit von IK10 gemäß EN...			
Target Release: PV_EL_GEN_2.20.0		Scope: System			
BL_OBB_40.00					
Allocation: DIGITAL INTERLOCKING (DIL)		Verification: System test			
DIGITAL FIELD ELEMENTS (DFE)		See also: ==> [EL4-SSS-REQ-00773] Vibrations and Shocks - FES-S, FES-M - Function			
		==> [EL4-SSS-REQ-00774] Vibrations and Shocks - FES Components - Function			
Issues/Decisions:					

EL4-DIL-SSS-REQ-00815		Vibrations and Shocks - FES-S, FES-M - Function		(Draft)	
The supplier shall verify the vibration and shock resistance of the [FES].					
Comments:					
Category:	TVR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_182] 6.1.0-1 Schwingungen und Stöße dürfen zu keiner Funktionsbeeinträchtigung einer Feldelementstation f...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
	DIGITAL FIELD ELEMENTS (DFE)				
Issues/Decisions:					

EL4-DIL-SSS-REQ-00816		Vibrations and Shocks - FES Components		(Draft)	
The supplier shall verify the vibration and shock resistance of the [FES] components.					
Comments:					
Category: TVR		Priority: Mandatory		Origin: ==> [EL4 PH] [03_184] 6.1.0-3 Schwingungen und Stöße dürfen zu keiner Funktionsbeeinträchtigung eines Teilsystems führen. ... System	
Originator: Customer		Origin:			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:			
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)		Verification: System test			
Issues/Decisions:		See also:			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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3.1.3.5 ENVIRONMENTAL PROTECTION

EL4-DIL-SSS-REQ-00776 Environmental Protection - Certification (Draft)	
The [DIL]'s supplier shall provide certification according to EN ISO 14001 or EMAS.	
<i>Comments:</i>	
Category: ECR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [03_189] 7.1.0-1 Der AN des DSTW muss eine Zertifizierung gemäß EN ISO 14001 oder gemäß EMAS nachweisen.
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00777 Environmental Protection - Hazardous Materials (Draft)	
The supplier shall use only materials that are not classified as hazardous materials in the event of disposal according to the current legal situation. The supplier shall request for the written consent of the customer, if the use of hazardous materials is unavoidable.	
<i>Comments:</i>	
Category: ECR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [03_191] 7.2.0-1 Es dürfen nur solche Materialien verwendet werden, die aufgrund der aktuellen Gesetzeslage i...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00778 Environmental Protection - Materials in Case of Fire (Draft)	
The supplier shall use only materials that do not release any aggressive or toxic compounds in case of fire. The supplier shall agree exceptions with the customer.	
<i>Comments:</i>	
Category: ECR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [03_193] 7.3.0-1 Es dürfen nur solche Materialien verwendet werden, die im Brandfall keine aggressiven und gi...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00779 Environmental Protection - Disposal Concept (Draft)	
The supplier shall create a disposal concept, agree it with the customer and implement it.	
<i>Comments:</i>	
<i>Category:</i> ECR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_195] 7.4.0-1 Der AN muss ein Entsorgungskonzept erstellen, mit dem AG abstimmen und umsetzen. System <i>Scope:</i> <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

3.1.3.6 PROTECTION RATING

EL4-DIL-SSS-REQ-00780 Protection Rating - DIL-CU Components (Draft)	
The components of the [DIL-CU] shall at least meet the requirements of EN 60529 protection rating [IP]20.	
<i>Comments:</i>	
<i>Category:</i> ECR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_198] 8.1.0-1 Die Komponenten der DSTW-ZE müssen mindestens die Anforderungen gemäß EN 60529 Schutzgrad IP... System <i>Scope:</i> <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00781 Protection Rating - DC Requirements (Draft)	
Requirements from [20 Data centre specifications] shall be met with regard to the degree of protection.	
<i>Comments:</i>	
<i>Category:</i> ECR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_199] 8.1.0-2 Anforderungen aus dem <20 Lastenheft Rechenzentrum> 1 müssen in Bezug auf den Schutzgrad ein... System <i>Scope:</i> <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00782		Protection Rating - Degree of Protection		(Draft)
The components for use in the [FES] shall at least meet the requirements of EN 60529 degree of protection [IP]20.				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_201] 8.2.0-1 Die Komponenten für den Einsatz in der FES müssen mindestens die Anforderungen gemäß EN 6052...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00783		Protection Rating - Outdoor Elements		(Draft)
The housing for the components of the outdoor system shall at least meet the requirements according to [EN 60529], degree of protection [IP]65.				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_203] 8.3.0-1 Die Einhausung für die Komponenten der Außenanlage muss mindestens die Anforderungen gemäß E...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

3.1.3.7 EMPLOYEE PROTECTION

3.1.3.7.1 REGULATIONS FOR EMPLOYEE PROTECTION

EL4-DIL-SSS-REQ-00784		Employee Protection - Regulations		(Draft)
The supplier shall observe all regulations for employee protection applicable in the customer's area, in particular [ASchG], [EisbAV], [AVO] Verkehr, [AM-VO], [KennV]. This applies to the construction phase, the planning phase and, in general, to the operation of the [DIL].				
Comments:				
Category:	PHR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_206] 9.1.0-1 Es müssen alle im Bereich des AG gültigen Vorschriften zum Arbeitnehmerschutz eingehalten we...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00785 Employee Protection - Preventive Services/Forces (Draft)	
All products supplied by the supplier shall meet the requirements of the preventive services or preventive forces. To this end, the supplier shall involve the preventive services or preventive forces throughout the entire life cycle and coordinate with them.	
<i>Comments:</i>	
<i>Category:</i> PHR <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_207] 9.1.0-2 Alle vom AN gelieferten Produkte müssen den Vorgaben der Präventivdienste bzw. Präventivkräf...
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

3.1.3.8 FIRE PROTECTION

EL4-DIL-SSS-REQ-00786 Employee Protection - Fire Protection - General (Draft)	
The supplier shall comply with all fire protection regulations applicable in the area of the customer. This applies both to the construction phase and, in general, to the operation of the [DIL].	
<i>Comments:</i>	
<i>Category:</i> PER <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_210] 10.1.0-1 Es müssen alle im Bereich des AG gültigen Vorschriften zum Brandschutz eingehalten werden. ...
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00787 Employee Protection - Fire Protection - Regulations Tunnel (Draft)	
The supplier shall take into account in particular the fire protection regulations for tunnels.	
<i>Comments:</i>	
<i>Category:</i> PER <i>Originator:</i> Customer	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_211] 10.1.0-2 Insbesondere müssen die Brandschutzverordnungen für Tunnel berücksichtigt werden.
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00788 Employee Protection - Fire Protection - Existing (Draft) Fire Protection Concept			
If parts of the system are installed in existing buildings, the supplier shall take into account an existing fire protection concept and, if necessary, adapt and update it.			
<i>Comments:</i>			
<i>Category:</i>	PER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_212] 10.1.0-3 Sofern Teile der Anlage in bestehende Gebäude eingebaut werden, muss ein vorhandenes Brands...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00789 Employee Protection - Fire Protection - Standard (Draft)			
All systems and devices installed in the track field shall comply with the fire protection standard EN 13501-2.			
<i>Comments:</i>			
<i>Category:</i>	PER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_213] 10.1.0-4 Sämtliche im Gleisfeld installierten Anlagen und Geräte müssen die Brandschutznorm EN 13501...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

3.1.3.9 SYSTEM PROTECTION

EL4-DIL-SSS-REQ-00790 Employee Protection - System Protection - (Draft) Concept			
The supplier shall submit a concept for protective measures against access to and manipulation of the system by unauthorised persons and agreed it with the customer.			
<i>Comments:</i>			
<i>Category:</i>	SER	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [03_216] 11.1.0-1 Es muss ein Konzept für Schutzmaßnahmen gegen Zugang zum System und Manipulation im System ...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00802		Employee Protection - System Protection - Customer Regulations		(Draft)	
The supplier shall observe the customer's applicable regulations in this regard.					
Comments:					
Category:	SER	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [03_216] 11.1.0-1 Es muss ein Konzept für Schutzmaßnahmen gegen Zugang zum System und Manipulation im System ...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
	DIGITAL FIELD ELEMENTS (DFE)				
Issues/Decisions:					

EL4-DIL-SSS-REQ-00803		Employee Protection - System Protection - Measures		(Draft)	
The protective measures shall take into account the installation of the [DIL] subsystems and system components in					
1. safety-related operating rooms,					
2. rooms with mixed functions, and					
3. outdoor areas.					
Comments:					
Category:		SER		Priority:	
Originator:		Customer		Mandatory	
Target Release:		PV_EL_GEN_2.20.0		Origin:	
Allocation:		BL_OBB_40.00		==> [EL4 PH] [03_216] 11.1.0-1 Es muss ein Konzept für Schutzmaßnahmen gegen Zugang zum System und Manipulation im System ...	
		DIGITAL INTERLOCKING (DIL)		Scope:	
		DIGITAL FIELD ELEMENTS (DFE)		System	
				Verification:	
				System test	
				See also:	
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.1.4 EULYNX IMPLEMENTATION

3.1.4.1 IMPLEMENTATION OF EULYNX REQUIREMENTS

EL4-DIL-SSS-REQ-00736		EULYNX - Specifications and Interfaces		(Draft)			
The supplier shall realize the [DIL] on the basis of the [EULYNX] architecture and the interfaces specified in the [EULYNX] specifications, as well as with existing interfaces based on [X25oRaSTA] to the existing [RBC], [OMC], [SCWS]/[SCWS-S] peripheral systems and neighbour interlockings [NIL].							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_98] 2.3.0-1 Das DSTW muss auf Basis der EULYNX-Architektur und den in EULYNX spezifizierten Schnittstell...	
Target Release:		PV_EL_GEN_2.20.0		Scope:		==> [EL4 PH] [03_256] 13.1.0-1 Alle beschriebenen EULYNX-Teilsysteme müssen entsprechend der für sie geltenden EULYNX-Spez...	
Allocation:		BL_OBB_40.00		Verification:		System test	
		DIGITAL INTERLOCKING (DIL)		See also:			
		ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)					
		DIGITAL FIELD ELEMENTS (DFE)					
Issues/Decisions:							

EL4-DIL-SSS-REQ-00804		EULYNX - Deviations		(In negotiation)	
The supplier shall agree deviations from the [EULYNX] specification with the customer and shall have them approved by the customer.					
Comments: == OPEN POINT == This is a process requirement.					
Category:	MCR	Priority:	Mandatory	==> [EL4 PH] [03_256] 13.1.0-1 Alle beschriebenen EULYNX-Teilsysteme müssen entsprechend der für sie geltenden EULYNX-Spez...	
Originator:	Customer	Origin:			
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: See also:	System test		
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00793		EULYNX - Conflicts		(In negotiation)
If a requirement from the [EULYNX] specification is in conflict with a requirement of this specification, The supplier shall resolve the deviation in consultation with the customer.				
Comments: == OPEN POINT == This is a process requirement.				
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_257] 13.1.0-2 Steht eine Anforderung aus der EULYNX-Spezifikation im Gegensatz zu einer Vorgabe dieses La...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

3.1.4.2 BASIC DATA IDENTIFIER

The parameters required at the "Basic Data Identifier" for the issue of the certificate are defined in [04 Security Requirements Specification].

EL4-DIL-SSS-REQ-00794		EULYNX - Basic Data Identifier - Compliance		(Draft)			
A "Basic Data Identifier" is to be implemented for each [EULYNX] subsystem in accordance with the [EULYNX] requirements.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_259] 13.2.0-1 Es ist für jedes EULYNX-Teilsystem ein Basic Data Identifier entsprechend der EULYNX-Anford...	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00		Verification:		System test	
		DIGITAL INTERLOCKING (DIL)		See also:			
		VIRTUAL INTERLOCKING (VIL)					
		DIGITAL FIELD ELEMENTS (DFE)					
Issues/Decisions:							

EL4-DIL-SSS-REQ-00795		EULYNX - Basic Data Identifier - Subsystem		(Draft)			
The "Basic Data Identifier" shall be physically identical for all subsystems.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [03_260] 13.2.0-2 Der Basic Data Identifier muss für alle Teilsystem physisch gleichartig ausgeführt werden. System	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)		Verification:		System test	
				See also:			
Issues/Decisions:							

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00796 EULYNX - Basic Data Identifier - Security (Draft)	
The "Basic Data Identifier" shall fulfil the security requirements.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_261] 13.2.0-3 Der Basic Data Identifier muss die Security-Anforderungen erfüllen. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00797 EULYNX - Basic Data Identifier - Initialisation (Draft)	
The "Basic Data Identifier" shall contain all the data necessary for automatic initialisation.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_262] 13.2.0-4 Der Basic Data Identifier muss alle notwendigen Daten für eine automatische Initialisierung... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00798 EULYNX - Basic Data Identifier - Retainment (Draft)	
The unique initial configuration (identifier, initial encryption) shall also be retained when the [OC] is replaced.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_264] 13.2.0-6 Die eindeutige Initialkonfiguration (Kennung, Initialverschlüsselung) muss auch bei Tausch... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00799 EULYNX - Basic Data Identifier - Non-Volatility (Draft)	
The initial configuration shall be non-volatile.	
Comments:	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [03_265] 13.2.0-7 Die Initialkonfiguration muss unverlierbar sein.
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
Issues/Decisions:	

3.1.4.3 ASYNCHRONOUS PRELOADING

EL4-DIL-SSS-REQ-00805 EULYNX - Asynchronous Preloading - DDSP (In negotiation)	
The [DDSP] peripheral system shall support the asynchronous preloading function according to [SMI].	
Comments: == OPEN POINT == Das ist ein Requirement.	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [03_267] 13.3.0-1 Das Umsystem DDSP unterstützt die Funktion Asynchronous Preloading gemäß SMI.
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00800 EULYNX - Asynchronous Preloading - Field Elements (Draft)	
All [EULYNX] field element subsystems shall implement the asynchronous preloading function according to [SMI].	
Comments:	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [03_268] 13.3.0-2 Alle EULYNX-Feldelement-Teilsysteme müssen die Funktion Asynchronous Preloading gemäß SMI u...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
Issues/Decisions:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.1.4.4 OUTDOOR SYSTEM START-UP

EL4-DIL-SSS-REQ-00801		EULYNX - Outdoor System Start-up		(Draft)
The object controllers for the outdoor system shall start up in accordance with the [EULYNX] specification that applies to them.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_270] 13.4.0-1 Die Objektcontroller der Außenanlagen müssen gemäß der für sie geltenden EULYNX-Spezifikati...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test	
	DIGITAL FIELD ELEMENTS (DFE)	See also:		
Issues/Decisions:				

3.2 D2.5 ELEKTRA 4.0 (DIL-CU)

3.2.1 ARCHITECTURE

3.2.1.1 GENERAL

The **Central Unit** of the Digital Interlocking **[DIL-CU]** consists of

- the Virtual Interlocking [VIL] - that is the safety logic in software, and
- the Server Hardware.

An instance of the safety logic on a Data Center [DC] server is called **[VIL] instance**.

The implementation of the safety logic in separate [VIL]s is necessary because

- an independent safety case of individual [VIL]s is achieved,
- standardisation of the interface between neighbour [VIL]s ([X25oRaSTA]/[SCI-ILS]) is required, and
- flexibility in adapting the feature set of the generic application shall be ensured.

The logical system border of the [DIL-CU] is shown in the following figure:

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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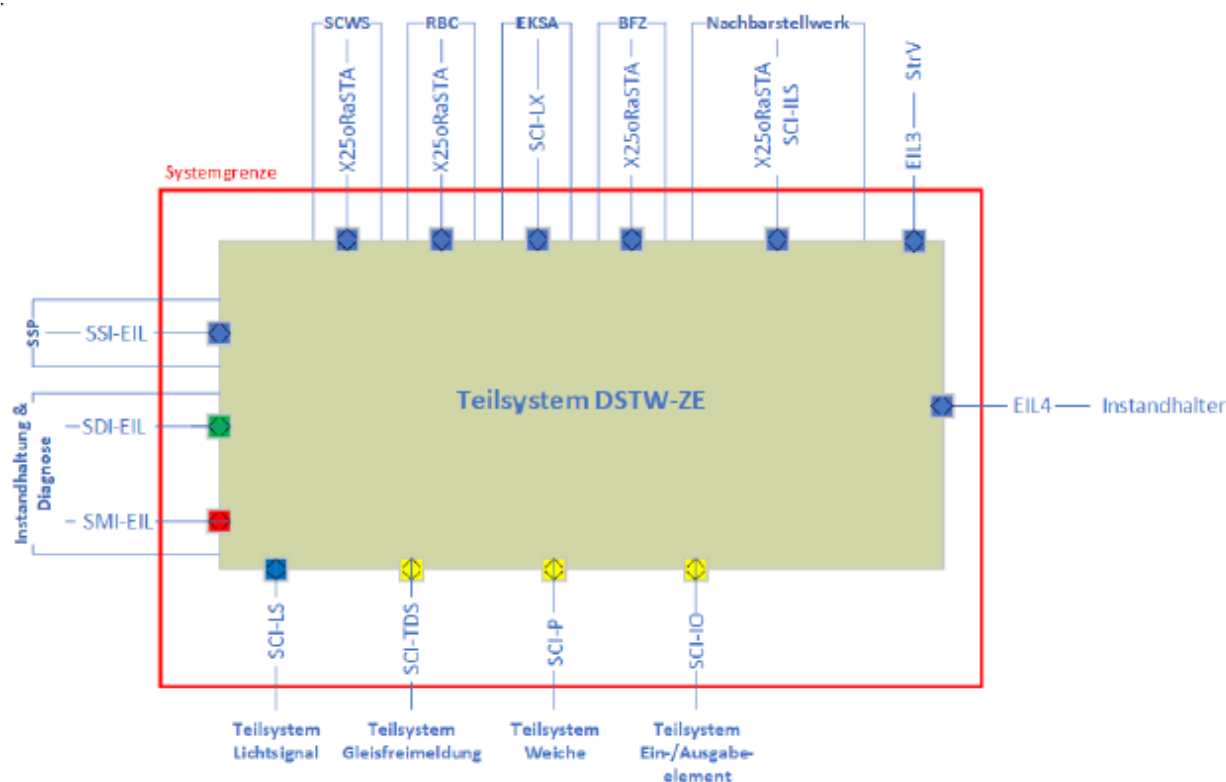


Figure 2: DI-CU: Logical System Border

== **TODO** ==

Update this picture.

EL4-DIL-SSS-REQ-00437 Architecture - Number of [VIL] Instances per Server (Draft)			
The [DIL] underlying safety concept and the [TAS Platform] mechanisms shall enable the simultaneous operation of at least eight [VIL] instances per server (one redundancy instance for each [VIL]).			
Comments:		Per server only [VIL]s of the same supplier are operated.	
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_24] 2.2.0-1 Das VSTW muss so beschaffen sein, dass mindestens acht VSTWs gleichzeitig auf einem Server b...
Target Release:	PV_EL_GEN_2.20.0	Scope:	==> [EL4 PH] [05_25] 2.2.0-2 Auf einem Server werden nur VSTW eines Lieferanten betrieben. System
Allocation:	BL_OBB_40.00	Verification:	System test
	DIGITAL INTERLOCKING (DIL)	See also:	
	INTEGRATED PLATFORM		
	SIGNALLING (IPS)		
	IPS Base		
Issues/Decisions:			

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00610		Architecture - Number of OCs		(Draft)			
The [DIL] shall support per [VIL] system controlling up to 1600 [OC]s.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [05_26] 2.2.0-3 Ein VSTW muss 1 bis 1600 Objectcontroller (OC) steuern können.	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00		Verification:		Subit test	
		DIGITAL INTERLOCKING (DIL)		See also:			
		VIRTUAL INTERLOCKING (VIL)					
		ELEKTRA Generic System					
		ELEKTRA Application					
		EC-MUX					
Issues/Decisions:							

3.2.1.2 [VIL] REDUNDANCY

The [ELEKTRA] 4.0 system ([DIL-CU]) is based on an "N times M out of M Redundancy" (NxMooM, [NMR]) concept, covering both the safety and the availability aspect:

- The required **Safety** is completely ensured by a 2oo2 system (= **Computing Node** [CN]). A single [CN] represents the basic form of a [VIL] delivering the complete functionality without redundancy.
- The required **Availability** of the system is ensured by the corresponding amount (N) of redundant [CN]s, which then form a **Computing Group** [CG].

The [ELEKTRA] 4.0 system architecture uses the [NMR] concept in the following variants:

- Basic (non-redundant) variant: 2x2oo2 (basic variant)
- Geo-redundant variant: 4x2oo2.

To maximize the availability the [VIL] redundancy instances are evenly distributed across the servers and data centre locations, in so-called **Availability Zones**.

EL4-DIL-SSS-REQ-00439		VIL Redundancy - Failure Detection and Recovery		(Draft)	
Each [VIL] redundancy instance shall detect the failure of other redundancy instances automatically and carry out any necessary recovery of the affected redundancy instances automatically.					
Comments:		This is a [TAS-PF] topic.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [05_32] 2.3.0-5 Redundante VSTW-Instanzen müssen den Ausfall eines oder mehrerer Redundanzpartner automatisch...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
	VIRTUAL INTERLOCKING (VIL)				
	ELEKTRA Generic System				
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00441		VIL Redundancy - Configuration		(Draft)	
All [VIL] redundancy instances shall be operated with identical configuration (generic platform, generic application and station data) ensured by technical measures.					
<u>Exception:</u> Short-term differences due to configuration changes resulting from updates or upgrades.					
Comments:		Identical configuration means identical assignment of [OC]s, [OMC]s, [RBC]s, [SCWS]s via the corresponding interfaces of the architecture image (e.g. ensuring identical hash values).			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [05_36] 2.3.0-9 Alle redundanten VSTW-Partner müssen mit identischer Konfiguration (generische Plattform, ge... ==> [EL4 PH] [05_41] 2.3.0-14 Die identische Konfiguration redundanter VSTW-Partner muss vom AN technisch sichergestellt ... ==> [EL4 PH] [05_42] 2.3.0-15 Identische Konfiguration bedeutet identische Zuordnung von OCs, BFZen, RBCs, SCWS über die ...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System CAE Data Tools INTEGRATED PLATFORM SIGNALLING (IPS) IPS Base	Verification:	System test		
		See also:			
Issues/Decisions:					

EL4-DIL-SSS-REQ-00442		VIL Redundancy - Safe Processing		(Draft)	
Each [VIL] instance shall autonomously provide safe processing, independent of whether other instances are running or not.					
Comments:		Operation without a redundancy instance e.g. makes it possible to carry out a software change or upgrade on a [VIL] that is not in operation, while train operation continues on the [VIL] that is still in operation.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 LH D2] [] ==> [EL4 PH] [05_37] 2.3.0-10 Während Wartungsarbeiten (Softwareupdates, Projektierungsänderungen, Hardwaretausch) müssen... ==> [EL4 PH] [05_38] 2.3.0-11 Der Betrieb ohne Redundanzpartner erlaubt es, auf einem nicht in Betrieb befindlichen VSTW ... ==> [EL4 PH] [05_40] 2.3.0-13 Das gewünschte Vorgehen zum Update, Änderungen der Projektierung ist in der Abbildung in ID...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
	VIRTUAL INTERLOCKING (VIL)				
	ELEKTRA Generic System				
Issues/Decisions:					

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EL4-DIL-SSS-REQ-00217		VIL Redundancy - Isolation of VIL Instances		(Draft)
A failure of a [VIL] instance shall not cause a failure of any other [VIL] instance running on the same server.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:		
Target Release:		Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:		
	INTEGRATED PLATFORM	See also:		
	SIGNALLING (IPS)			
Issues/Decisions:				

The desired update procedure, including [SW] changes or data preparation changes, is shown in the figure below. The aim is to maintain train operations while one of the redundancy instance is shut down and updated.

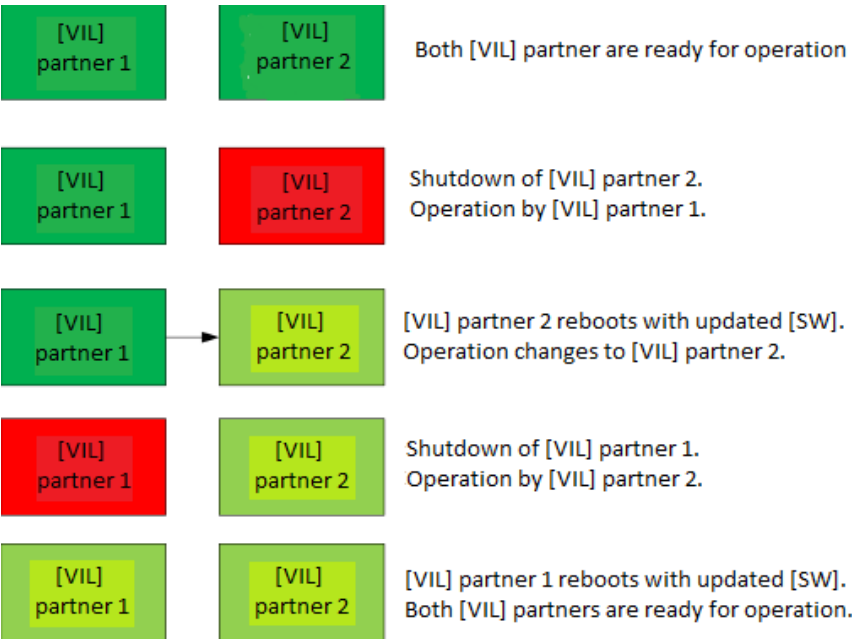


Figure 3: Software Change

== **TODO** ==
Update this picture.

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EL4-DIL-SSS-REQ-00443		VIL Redundancy - Availability		(Draft)	
The [VIL] redundancy instances shall be used exclusively for increasing the availability.					
Comments:		This means e.g. that the redundancy cannot be used to achieve the safety objectives (see also [EL4-DIL-SSS-REQ-00442]).			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [05_43] 2.3.0-16 Die redundanten VSTW-Partner dienen ausschliesslich der Verfügbarkeitserhöhung. ==> [EL4 PH] [05_44] 2.3.0-17 Die redundanten VSTW-Partner dürfen keine Voraussetzung für das Erreichen der Sicherheitszi... ==> [EL4 SSS]		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test		
		See also:	==> [EL4-SSS-REQ-00442] VIL Redundancy - Safe Processing		
Issues/Decisions:					

3.2.1.3 [DIL] VARIANTS

3.2.1.3.1 GENERAL REQUIREMENTS

EL4-DIL-SSS-REQ-00448		DIL Variants - COTS Hardware		(Draft)	
The [DIL] servers shall be equipped with freely available [COTS] hardware.					
Comments:					
Category: FUR		Priority: Mandatory		==> [EL4 PH] [05_52] 3.1.0-3 Die Server der DSTW-ZE müssen mit am Markt frei verfügbarer Hardware ausgerüstet werden. System	
Originator: Customer		Origin:			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:			
Allocation: DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) IPS Base		Verification: System test See also:			
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00451		DIL Variants - Hot-hot Redundancy		(Draft)	
The [DIL] system shall provide a hot-hot redundancy for the [VIL] applications by executing them in parallel on several servers. Thus, in case of a failure of a [VIL] instance, a [VIL] redundant instance shall continue processing and perform the service functions without interruption or data loss (= seamless operation).					
Comments:		•Seamless operation implies that all states are taken over by the redundancy partner. •Specific examples for seamless operation: •Signals shall not be set to stop. •There shall be no need to reset routes.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [05_28] 2.3.0-1 In der Variante Virtualisierung müssen alle VSTW in mindestens zwei redundanten Instanzen pa... ==> [EL4 PH] [05_29] 2.3.0-2 In den Varianten Georedundanz und Hardwareunabhängigkeit müssen alle VSTW in mindestens drei... ==> [EL4 PH] [05_56] 3.2.0-3 Bei Ausfall des primären VSTW-Partner muss ein redundanter VSTW-Partner den Betrieb weiterfü... ==> [EL4 PH] [05_57] 3.2.0-4 Die Übernahme durch den redundanten VSTW-Partner nach Ausfall des primären VSTW-Partners mus... ==> [EL4 PH] [05_58] 3.2.0-5 Bei einem Ausfall des primären VSTW-Partners muss der übernehmende Redundanzpartner den Zust...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
	VIRTUAL INTERLOCKING (VIL)				
	ELEKTRA Generic System				
Issues/Decisions:					

Hitachi Rail – Internal

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EL4-DIL-SSS-REQ-00453		DIL Variants - SW Updates		(Draft)	
Software updates shall be carried out without interrupting train operations (by using the redundant instance).					
Exception: Software updates with safety impact.					
Comments:					
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [05_55] 3.2.0-2 Softwareupdates (ausgenommen Anpassungen mit Safety-relevanz) müssen ohne Unterbrechnung des...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) IPS Base IPS Infra ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)	Verification:	System test		
		See also:			
Issues/Decisions:					

3.2.1.3.2 BASIC VIRTUALISATION (VI-1)

In the variant "Basic Virtualisation" a [DC] is planned in which (locally) [VIL] redundant instances ensure availability.

EL4-DIL-SSS-REQ-00450		DIL Variants - Basic Virtualisation		(Draft)	
In the variant "Basic Virtualisation" it shall be possible to run all [VIL]s in at least two redundant instances in parallel on different servers at a local [DC] location, so that the failure of one instance does not result in the loss of the overall function.					
Comments:					
Category: FUR		Priority: Mandatory		==> [EL4 PH] [05_28] 2.3.0-1 In der Variante Virtualisierung müssen alle VSTW in mindestens zwei redundanten Instanzen pa... ==> [EL4 PH] [05_30] 2.3.0-3 VSTW der Variante Virtualisierung müssen an einem lokalen Rechenzentrums-Standort ausführbar... ==> [EL4 PH] [05_54] 3.2.0-1 Für jedes VSTW dieser Ausprägung muss am selben Standort ein Redundanzpartner realisiert wer...	
Originator: Customer		Origin:			
Target Release: PV_EL_GEN_2.20.0		Scope: System			
Allocation: DIGITAL INTERLOCKING (DIL)		Verification: System test			
VIRTUAL INTERLOCKING (VIL)		See also:			
ELEKTRA Generic System					
INTEGRATED PLATFORM					
SIGNALLING (IPS)					
IPS Base					
Issues/Decisions:					

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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3.2.1.3.3 ENHANCEMENT GEO-REDUNDANCY (VI-2)

In the variant "Geo-Redundancy" it is planned to implement several geo-redundant [DC]s on which the [VIL] instances are executed.

EL4-DIL-SSS-REQ-00456		DIL Variants - Geo-Redundancy - Different DC Locations		(Draft)
In the variant "Geo-Redundancy" the [VIL] redundant instances shall be able to run in parallel at up to three geo-redundant [DC] locations.				
Comments:		The third redundancy is necessary to solve the [Split Brain] problem.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_31] 2.3.0-4 Die redundanten Instanzen VSTW der Varianten Georedundanz und Hardwareunabhängigkeit müssen ...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) IPS Base	Verification:	System test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00457		DIL Variants - Geo-Redundancy - Different Servers		(Draft)
In the variant "Geo-Redundancy" it shall be possible to execute all [VIL]s in at least three redundant instances in parallel on different servers, so that the failure of at least one redundant instance does not result in the loss of the overall function.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_29] 2.3.0-2 In den Varianten Georedundanz und Hardwareunabhängigkeit müssen alle VSTW in mindestens drei...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	ELEKTRA Generic System			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	IPS Base			
Issues/Decisions:				

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00699		DIL Variants - Geo-Redundancy - Splitting of Transport Channels		(Draft)
In the variant "Geo-Redundancy" the [VIL] shall handle the failure of at least one redundant instance of the same [DC] site by splitting the two [RaSTA] transport channels between the sites.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_29] 2.3.0-2 In den Varianten Georedundanz und Hardwareunabhängigkeit müssen alle VSTW in mindestens drei...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test	
	BL_OBB_40.00	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	ELEKTRA Generic System			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00700		DIL Variants - Geo-Redundancy - Manual Switching of Transport Channels		(In negotiation)
In the variant "Geo-Redundancy" the [VIL] shall support the manual switchover of the transport channels via [ADM].				
Comments: == OPEN POINT == Does this requirement still makes sense? In the meantime the customer requests an <u>automatic switchover</u> .				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_29] 2.3.0-2 In den Varianten Georedundanz und Hardwareunabhängigkeit müssen alle VSTW in mindestens drei...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	ELEKTRA Generic System			
	ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)			
Issues/Decisions:				

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00458		DIL Variants - Geo-Redundancy - Seamless Operation		(Draft)
In the variant "Geo-Redundancy", in case of a failure at the [DC] location, the geo-[VIL] redundant instances at the geo-redundant [DC] location shall continue to operate without interruption or data loss (seamless operation).				
Comments:		•Seamless operation implies that all states are taken over by the redundant instance. •Specific examples for seamless operation: •Signals shall not be set to stop. •There shall be no need to reset routes.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_60] 3.3.0-1 Bei Ausfall des Rechenzentrumsstandort, an dem der primäre VSTW-Partner und der lokale Redun... ==> [EL4 PH] [05_61] 3.3.0-2 Die Übernahme durch den Redundanzpartner nach Ausfall des primären VSTW-Partners muss erfolg... ==> [EL4 PH] [05_62] 3.3.0-3 Bei einem Ausfall des primären VSTW-Partners muss der übernehmende Redundanzpartner den Zust...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) IPS Base	Verification:	System test	
		See also:		
Issues/Decisions:				

3.2.1.3.4 ENHANCEMENT INDEPENDENCE FROM SERVER HARDWARE (VI-3)

In the variant "Hardware Independence" it is planned to gain independence from a special server hardware.

The variant "Hardware Independence" will be released in a later baseline.

EL4-DIL-SSS-REQ-00462		DIL Variants - Hardware Independence - Safety Case		(Descoped)
In the variant "Hardware Independence" a change of the server hardware shall not require a change of the [VIL]'s safety case.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_64] 3.4.0-1 Eine Änderung der Hardware, auf dem ein VSTW der Erweiterung Hardwareunabhängigkeit ausgefüh...	
Target Release:		Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) IPS Base	Verification:	System test	
		See also:		
Issues/Decisions:				

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00463		DIL Variants - Hardware Independence - Customer's Choice		(Descoped)	
In the variant "Hardware Independence" the [VIL] shall be executable on hardware selected and provided by the customer.					
Comments:					
Category: FUR		Priority: Mandatory		==> [EL4 PH] [05_65] 3.4.0-2 Ein VSTW der Erweiterung Hardwareunabhängigkeit muss auf einer vom Auftraggeber ausgewählten... System System test	
Originator: Customer		Origin:			
Target Release:		Scope:			
Allocation:		Verification:			
		See also:			
DIGITAL INTERLOCKING (DIL)					
VIRTUAL INTERLOCKING (VIL)					
ELEKTRA Generic System					
INTEGRATED PLATFORM					
SIGNALLING (IPS)					
IPS Base					
Issues/Decisions:					

3.2.1.4 ACTIVATION OF SOFTWARE UPDATES

According the [DDSP] concept, the [DDSP] offers the information to [VIL] concerning a software update that can be activated.

EL4-DIL-SSS-REQ-00695		Activation of Software Updates - Command from OMC		(Draft)	
The [DIL] shall execute the activation command received from the [OMC] for a software update received from the [DDSP] for itself and the [OC]s belonging to it.					
Comments:		Received via [X25] main group 177 / control word 1 / bit 6.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [05_1285] 2.4.0-1 Das VSTW muss den von der BFZ empfangenen Aktivierungsbefehl für ein von der DDSP empfangene...		
			==> [EL4 PH] [05_1289] 2.4.0-5 Die Freigabe zur Aktivierung eines Software-Updates wird von der BFZ an das VSTW kommandiert...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
	BL_OBB_40.00				
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test		
	VIRTUAL INTERLOCKING (VIL)	See also:			
	ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)				
	DIGITAL FIELD ELEMENTS (DFE)				
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00696 Activation of Software Updates - Authorisation Check (Draft)	
The [DIL] shall check whether a received software update is authorised or not.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_1287] 2.4.0-3 Das VSTW muss überprüfen, ob ein empfangenes Software-Update zulässig ist. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00697 Activation of Software Updates - Message to OMC (Draft)	
The [DIL] shall inform the [OMC] when a software update is available.	
<i>Comments:</i> Sent via [X25] main group 57 / control word 1 / bit 4.	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_1288] 2.4.0-4 Das VSTW muss die BFZ über das Vorliegen eines Software-Update informieren. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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3.2.2 INTERFACES TO ADJACENT SYSTEMS

3.2.2.1 OMC

EL4-DIL-SSS-REQ-00464 Interface to OMC - Communication Protocol (Draft)			
For each [VIL] the [DIL] shall provide an interface to an [OMC] system using the communication protocol [X25 over RaSTA]. The interface shall be compliant with the following specifications:: •[LH X25 – Schnittstelle ÖBB] •[PH X25 – BFZ Schnittstelle ÖBB] •[LH – EBO2 / X25] •[Schnittstellenbeschreibung X25oIP]			
<i>Comments:</i>			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [05_68] 4.1.0-1 Es muss eine Umsystemschnittstelle zur BFZ entsprechend der folgenden Spezifikationen realis...
			==> [EL4 PH] [05_69] 4.1.0-2 Die Anbindung an die Bedienebene (BFZ) muss über X25oRaSTA realisiert werden.
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Verification:</i>	System test
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00465 Interface to OMC - "BFZ-Ausfallkonzept Phase 1 (Siemens)" (Draft)			
The [OMC] Interface shall support the [OMC] failure scenario "BFZ-Ausfallkonzept Phase 1 (Siemens)".			
<i>Comments:</i>			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [05_70] 4.1.0-3 Die Anbindung an die Bedienebene muss das BFZ-Ausfallkonzept Phase 1 beherrschen (Siemens).
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Verification:</i>	System test
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00466 Interface to OMC - "BFZ-Ausfallskonzept Phase 2 (Thales)" (Draft)			
The [OMC] Interface shall support the [OMC] failure scenario "BFZ-Ausfallskonzept Phase 2 (Thales)".			
<i>Comments:</i>			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [05_71] 4.1.0-4 Die Anbindung an die Bedienebene muss das BFZ-Ausfallskonzept Phase 2 beherrschen (Thales).
<i>Target Release:</i>	PV_EL_GEN_2.20.0	<i>Scope:</i>	System
<i>Allocation:</i>	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00467 Interface to OMC - EULYNX (Descoped)			
For each [VIL] the [DIL] shall provide an interface to an [OMC] system according to the following EULYNX specifications:			
•[Eu.Doc.50 - Interface Specification SCI-CC] •[Eu.Doc.49 - Requirement Specification SCI-CC]			
<i>Comments:</i>			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [05_72] 4.1.0-5 Es muss eine Umsystemschnittstelle zur BFZ entsprechend der folgenden Spezifikationen realisie... ==> [EL4 PH] [05_73] 4.1.0-6 Die Anbindung an die Bedienebene (BFZ) muss über SCI-CC realisiert werden.
<i>Target Release:</i>		<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

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3.2.2.2 RBC

EL4-DIL-SSS-REQ-00468 Interface to RBC - Communication Protocol (Draft)	
For each [VIL] the [DIL] shall provide an [RBC] interface using the communication protocol [X25 over RaSTA]. The interface shall be compliant with the following specifications: •[Österreichische Bundesbahnen - Generaldirektion - Sicherungsdienst - PH X25 - Schnittstelle] •[RAHMENAUSSCHREIBUNG - ETCS STRECKENAUSRÜSTUNG - D.1 Lastenheft]	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_75] 4.2.0-1 Es muss eine Umsystemschnittstelle zum RBC entsprechend der folgenden Spezifikationen realis... ==> [EL4 PH] [05_76] 4.2.0-2 Die Anbindung an die Radio Block Center (RBC) muss über X25oRaSTA realisiert werden. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00469 Interface to RBC - EULYNX (Descoped)	
For each [VIL] the [DIL] shall provide an [RBC] interface according to the following [EULYNX] specifications: •[Eu.Doc.47 - Requirement Specification for SCI-RBC] •[Eu.Doc.48 - Interface Specification SCI-RBC]	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_77] 4.2.0-3 Es muss eine Umsystemschnittstelle zum RBC entsprechend der folgenden Spezifikationen realis... ==> [EL4 PH] [05_78] 4.2.0-4 Die Anbindung an die Radio Block Center (RBC) muss über SCI-RBC realisiert werden. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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3.2.2.3 SCWS

EL4-DIL-SSS-REQ-00470 Interface to SCWS - Communication Protocol (Draft)	
For each [VIL] the [DIL] shall provide an interface to an [SCWS] system using the communication protocol [X25 over RaSTA]. The interface shall be compliant with the following specifications: •[LH X25 – Schnittstelle ÖBB] •[PH X25 – BFZ Schnittstelle ÖBB] •[LH – EBO2 / X25] •[Schnittstellenbeschreibung X25oIP]	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_80] 4.3.0-1 Es muss eine Umsystemschnittstelle zum SCWS entsprechend der folgenden Spezifikationen reali... ==> [EL4 PH] [05_81] 4.3.0-2 Die Anbindung an die Rottenwarnsysteme (SCWS) muss über X25oRaSTA realisiert werden. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00471 Interface to SCWS - EULYNX (Descoped)	
For each [VIL] the [DIL] shall provide an [SCWS] interface according to the following [EULYNX] specifications: •[Eu.Doc.49 - Requirement Specification SCI-CC] •[Eu.Doc.50 - Interface Specification SCI-CC]	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_82] 4.3.0-3 Es muss eine Umsystemschnittstelle zum SCWS entsprechend der folgenden Spezifikationen reali... ==> [EL4 PH] [05_83] 4.3.0-4 Die Anbindung an die Rottenwarnsysteme (SCWS) muss über SCI-CC realisiert werden <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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3.2.2.4 ADJACENT INTERLOCKING

EL4-DIL-SSS-REQ-00472 Interface to Adjacent IL - Adjacent VIL - Communication Protocol - Same Server (Draft)	
For each [VIL] the [DIL] shall provide an interface to adjacent [VIL]s <u>within the same server</u> using the communication protocol [X25 over RaSTA]. The interface shall be compliant with the following specifications: •[LH X25 – Schnittstelle ÖBB] •[Schnittstellenbeschreibung X25oIP]	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_85] 4.4.0-1 Es muss eine Umsystemschnittstelle zu Nachbarstellwerken entsprechend der folgenden Spezifik... ==> [EL4 PH] [05_86] 4.4.0-2 Die Kommunikation zwischen VSTW und benachbarten VSTW innerhalb eines Servers muss über X25o... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00733 Interface to Adjacent IL - Adjacent VIL - Communication Protocol - Same DC Location (Draft)	
For each [VIL] the [DIL] shall provide an interface to adjacent [VIL]s <u>within the same [DC] location</u> using the communication protocol [X25 over RaSTA]. The interface shall be compliant with the following specifications: •[LH X25 – Schnittstelle ÖBB] •[Schnittstellenbeschreibung X25oIP]V	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_85] 4.4.0-1 Es muss eine Umsystemschnittstelle zu Nachbarstellwerken entsprechend der folgenden Spezifik... ==> [EL4 PH] [05_87] 4.4.0-3 Die Kommunikation zwischen VSTW und benachbarten VSTW innerhalb eines RZ-Standort muss herst... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00734 Interface to Adjacent IL - Adjacent VIL - Communication Protocol - Different DC Locations		(Draft)
For each [VIL] the [DIL] shall provide an interface to adjacent [VIL]s <u>outside of the same [DC] location</u> using the communication protocol [X25 over RaSTA]. The interface shall be compliant with the following specifications: •[LH X25 – Schnittstelle ÖBB] •[Schnittstellenbeschreibung X25oIP]		
<i>Comments:</i>		
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_85] 4.4.0-1 Es muss eine Umsystemschnittstelle zu Nachbarstellwerken entsprechend der folgenden Spezifik... ==> [EL4 PH] [05_88] 4.4.0-4 Die Kommunikation zwischen VSTW und benachbarten VSTW außerhalb eines RZ-Standort muss herst... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>	
<i>Issues/Decisions:</i>		

EL4-DIL-SSS-REQ-00732 Interface to Adjacent IL - Any Adjacent IL - Communication Protocol		(Draft)
For each [VIL] the [DIL] shall provide an interface to any adjacent [IL] using the communication protocol [X25 over RaSTA]. The interface shall be compliant with the following specifications: •[LH X25 – Schnittstelle ÖBB] •[Schnittstellenbeschreibung X25oIP]		
<i>Comments:</i>		
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_85] 4.4.0-1 Es muss eine Umsystemschnittstelle zu Nachbarstellwerken entsprechend der folgenden Spezifik... ==> [EL4 PH] [05_89] 4.4.0-5 Die Kommunikation zwischen VSTW und einem benachbarten Stellwerk (STW) muss über X25oRaSTA e... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>	
<i>Issues/Decisions:</i>		

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00473		Interface to Adjacent IL - Converter		(Draft)			
If it is not possible to modify the existing adjacent [IL] and the existing [IL] does not have a native [X25] connection, the supplier shall implement a converter.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [05_90] 4.4.0-6 Ist eine Umrüstung des Bestands-Nachbarstellwerks nicht möglich und das Bestandsstellwerk we...	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application DIGITAL FIELD ELEMENTS (DFE) IFC-Block		Verification:		System test	
				See also:			
Issues/Decisions:							

EL4-DIL-SSS-REQ-00474		Interface to Adjacent IL - EULYNX		(Descoped)	
For each [VIL] the [DIL] shall provide an interface to adjacent [IL]s according to the following [EULYNX] specifications:					
•[Eu.Doc.41 – Requirement Specification for SCI-ILS]					
•[Eu.Doc.42 – Interface Specification for SCI-ILS]					
Comments:		This also includes the communication with a adjacent [IL] of a different railway operator (e.g. [DB-Netz[, [SŽ – Infrastruktura[, [SBB], [MÁV], [RFI]).			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [05_91] 4.4.0-7 Es muss eine Umsystemschnittstelle zu Nachbarstellwerken entsprechend der folgenden Spezifik... ==> [EL4 PH] [05_92] 4.4.0-8 Die Kommunikation zwischen VSTW und einem Nachbarstellwerk eines anderen Bahnbetreibers (zum...		
Target Release:		Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test		
		See also:			
Issues/Decisions:					

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00475		Interface to Adjacent IL - Technical Details		(In negotiation)
The technical details for the adaptation to a adjacent [EIL] or to a adjacent [IL] shall be clarified by the supplier with the respective [EIL]/[IL] manufacturer.				
Comments: == OPEN POINT == •Does [IL] here refer to Relay Interlockings? •This is a process requirement!				
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_93] 4.4.0-9 Die technischen Details zur Anpassung an ein Nachbar-ESTW bzw. Nachbar-STW müssen vom AN mit...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	ELEKTRA Application			
Issues/Decisions:				

3.2.2.5 MAINTENANCE AND DIAGNOSIS

EL4-DIL-SSS-REQ-00476		Maintenance and Diagnosis - Interface SDI-EIL		(Draft)
For each [VIL] the [DIL] shall provide a diagnosis interface [SDI-EIL] to the subsystem "Maintenance and Data Management" [DDSP] according to the following EULYNX specifications:				
•[Eu.Doc.94 - Interface specification SDI generic]				
•[Interface Specification SDI-EIL]				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_95] 4.5.0-1 Es muss eine Schnittstelle SDI-EIL zum Subsystem Maintenance and Data Management/DDSP realis...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	ELEKTRA Application			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	ADMINISTRATION, DIAGNOSTIC			
	AND MAINTENANCE-SERVICES			
	(ADM)			
Issues/Decisions:				

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EL4-DIL-SSS-REQ-00477		Maintenance and Diagnosis - Interface SMI-EIL		(Draft)
For each [VIL] the [DIL] shall provide a maintenance interface [SMI-EIL] to the subsystem "Maintenance and Data Management" [DDSP] according to the following EULYNX specifications:				
•[Eu.Doc.76 - Interface definition and specification SMI]				
•[Interface Specification SMI-EIL]				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_96] 4.5.0-2 Es muss eine Schnittstelle SMI-EIL zum Subsystem Maintenance and Data Management/DDSP realis...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	ELEKTRA Application			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	ADMINISTRATION, DIAGNOSTIC			
	AND MAINTENANCE-SERVICES			
	(ADM)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00478		Maintenance and Diagnosis - Interface EIL2		(Descoped)
For each [VIL] the [DIL] shall provide an interface [EIL2] to the subsystem "Maintenance and Data Management" [DDSP].				
Comments:		Interface [EIL2] is descoped.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_97] 4.5.0-3 Es muss eine Schnittstelle EIL2 zum Subsystem Maintenance and Data Management/DDSP realisier...	
Target Release:		Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)	Verification:	System test	
		See also:		
Issues/Decisions:				

3.2.2.6 POWER SUPPLY

The power supply [EIL3] is provided by the customer (see [Lastenheft Rechenzentrum]).

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00481 Power Supply - Support of DC Power Supply (Draft)	
The hardware of the [DIL-CU] shall be able to be operated on the power supply provided by the [DC].	
Comments: Further details see chapter [D4 Data Center].	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [05_102] 4.7.0-2 Die Hardware der DSTW-ZE muss an der vom Rechenzentrum zur Verfügung gestellten Stromversorg...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) IPS Base	Verification: System test See also:
Issues/Decisions:	

3.2.2.7 MAINTAINER

EL4-DIL-SSS-REQ-00482 Maintainer - Interface EIL4 (In negotiation)	
Each [VIL] shall provide an interface [EIL4] to the Maintainer.	
Comments: == OPEN POINT == Check EULYNX specification.	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [05_104] 4.8.0-1 Es muss eine Schnittstelle EIL4 realisiert werden.
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)	Verification: System test See also:
Issues/Decisions:	

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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3.2.3 SYSTEM STARTUP

3.2.3.1 STARTUP OF A [VIL] INSTANCE WITHOUT CONFIGURATION CHANGE

EL4-DIL-SSS-REQ-00655 Startup VIL Instance - No Configuration Change (In negotiation) - Automatic Start-up			
After automatic server shutdowns or when the power supply is restored after supply failures, the [VIL] instance shall initiate automatic start-up attempts.			
Comments: == OPEN POINT == Shall we define a maximum time for the re-integration?			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_121] 5.1.0-1 Nach automatischen Rechnerabschaltungen bzw. bei Spannungswiederkehr nach Versorgungsausfall...
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS)	Verification:	System test
See also:			
Issues/Decisions:			

EL4-DIL-SSS-REQ-00716 Startup VIL Instance - No Configuration Change (Draft) - Retrieving of States			
After a successful start-up the [VIL] shall retrieve all states assigned to the [VIL] and all redundant instances.			
Comments:			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_121] 5.1.0-1 Nach automatischen Rechnerabschaltungen bzw. bei Spannungswiederkehr nach Versorgungsausfall...
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS)	Verification:	System test
See also:			
Issues/Decisions:			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00656 Startup VIL Instance - No Configuration Change (Draft) - Necessary Data after Shutdown			
In case of an unplanned shutdown of a [VIL] instance, after the startup of a [VIL] without configuration changes, a [VIL] instance shall have at least the following data available:			
•Preparation data •Status data of the peripheral system •Status data of all redundancy partners •Current system data (subsidiary or caution signal, point machine or double key lock at a point) •Release status (single key lock status, double key lock status) •Current number of the registration of documented events •Last status of the locks (points, flat crossings, [EKW], [DKW], derailleurs, signals, key locks, goal locks, direction change only for outgoing direction) •Last status of the indication signs set •Last status of the auxiliary locks set •Transfer of data from [SCWS] computer •Last status of the train movement lock set (status always not bridged) •Last status of the [FLAB] entries			
<i>Comments:</i> <u>Note:</u> In this case, due to the applied "Hot-Hot Redundancy" strategy no data loss will occur and this requirement is implicitly fulfilled.			
<i>Category:</i> FUR <i>Originator:</i> Customer		<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [05_122] 5.1.0-2 Ein VSTW-Partner muss nach Inbetriebsetzung ohne Konfigurationsänderung mindestens über folg...	
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application INTEGRATED PLATFORM SIGNALLING (IPS) IPS Service		<i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>	
<i>Issues/Decisions:</i>			

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00657 Startup VIL Instance - No Configuration Change (Draft) - Message to OMC after Shutdown-			
In case of an unplanned shutdown of a [VIL] instance, a message shall be generated and sent to the [OMC].			
Comments:			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_123] 5.1.0-3 Bei ungeplanter Außerbetriebsetzung eines VSTW-Partners muss eine Meldung generiert und an d...
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test
		See also:	
Issues/Decisions:			

3.2.3.2 STARTUP OF ALL [VIL] INSTANCES WITHOUT CONFIGURATION CHANGE

EL4-DIL-SSS-REQ-00658 Startup all VIL Instances - No Configuration Change - Necessary Data after Shutdown - General (Draft)			
In case of an unplanned shutdown of <u>all</u> [VIL] instances, after the startup of a [VIL] without configuration change, each [VIL] instance shall have available at least the data according to [EL4-DIL-SSS-REQ-00656].			
Comments: Function according to ELEKTRA 2 baseline 29.20.			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_125] 5.2.0-1 Bei der Inbetriebnahme eines DSTW ohne Konfigurationsänderung (ungeplante Außerbetriebsetzun...
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test
		See also:	==> [EL4-SSS-REQ-00656] Startup VIL Instance - No Configuration Change - Necessary Data after Shutdown
Issues/Decisions:			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00730	Startup all VIL Instances - No Configuration Change - Default Values after Inconsistencies - General	(Draft)
In case of data inconsistencies after the startup of a [VIL] without configuration change, the [VIL] shall retrieve the following default values: •Current system data (subsidiary or caution signal): subsidiary signal •Current system data (point machine or double key lock at a point): point machine •Release status (single key lock, double key lock): [NADA] •Current number of the registration of documented events: 0 •Last status of locks: locked •Last status of indication signs set: indication signs set and not acknowledged •Last status of [SCWS-S]: troop set and not acknowledged •Last status of train movement lock: set, not bridged •For [FLAB] entries the procedure for long-term failure shall be followed		
Comments: Function according to ELEKTRA 2 baseline 29.20.		
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Priority: Mandatory Origin: ==> [EL4 PH] [05_125] 5.2.0-1 Bei der Inbetriebnahme eines DSTW ohne Konfigurationsänderung (ungeplante Außerbetriebsetzun... Scope: System Verification: System test See also:	
Issues/Decisions:		

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00659 Startup all VIL Instances - No Configuration Change - Default Values after Inconsistencies - Assent (Draft)	
In case of an unplanned shutdown of <u>all</u> [VIL] instances, after the startup of a [VIL] without configuration change, concerning "Assent" the [VIL] shall retrieve the following default values: •No assent given for travel in own area •No assent request requested for travel in neighbour area •If the message "assent given for travel in neighbour area" is present or this message is received as part of a general request from an [X25] interface, this assent is to be marked as "unusable". •No assent given for flank protection •No assent requested for flank protection	
Comments: Function according to ELEKTRA 2 baseline 29.20.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Priority: Mandatory Origin: ==> [EL4 PH] [05_126] 5.2.0-2 Bei der Inbetriebnahme eines DSTW ohne Konfigurationsänderung (ungeplante Außerbetriebsetzun... Scope: System Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00660 Startup all VIL Instances - No Configuration Change - Necessary Data after Shutdown - Line Block (Draft)	
Concerning "Line Block", after a short-term shutdown, the [DIL] shall retrieve the values according to chapter [Line Block - Procedure in case of Irregularities].	
Comments: Function according to ELEKTRA 2 baseline 29.20.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Priority: Mandatory Origin: ==> [EL4 PH] [05_127] 5.2.0-3 Für den Streckenblock sind die Angaben zu Daten nach einer kurzfristigen Außerbetriebsetzung... Scope: System Verification: System test See also:
Issues/Decisions:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.2.3.3 STARTUP OF A [VIL] INSTANCE WITH CONFIGURATION CHANGE

EL4-DIL-SSS-REQ-00661 Startup VIL Instance - Configuration Change - Necessary Data after long-term Shutdown (In negotiation)	
After commissioning of a [VIL] with configuration changes (planned decommissioning of a [VIL] instance) a [VIL] shall have available the following data: •Preparation data •Status data of the peripheral system •Status data of all redundancy partners •Current system data (subsidiary or caution signal, point machine or double key lock at a point).	
Comments: == OPEN POINT == Is this case possible at all? A computing node with changed station data cannot be integrated at all due to TAS Plf constraints.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application CAE Data Tools INTEGRATED PLATFORM SIGNALLING (IPS) IPS Service	Priority: Mandatory Origin: ==> [EL4 PH] [05_129] 5.3.0-1 Für den Fall Inbetriebsetzung eines VSTW mit Konfigurationsänderung (Außerbetriebsetzung ein... Scope: System Verification: System test See also:
Issues/Decisions:	

3.2.3.4 STARTUP OF ALL [VIL] INSTANCES WITH CONFIGURATION CHANGE

EL4-DIL-SSS-REQ-00662 Startup all VIL Instances - Configuration Change (Draft) - Necessary Data after long-term Shutdown	
In case of a shutdown of <u>all</u> [VIL] instances, after the startup of a [VIL] with configuration change, each [VIL] instance shall have available at least the data according to [EL4-DIL-SSS-REQ-00656].	
Comments: Function according to ELEKTRA 2 baseline 29.20.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Priority: Mandatory Origin: ==> [EL4 PH] [05_131] 5.4.0-1 Für den Fall einer Außerbetriebsetzung des VSTW (Außerbetriebsetzung aller VSTW-Partner): K... Scope: ==> [EL4 SSS] System Verification: System test See also: ==> [EL4-SSS-REQ-00656] Startup VIL Instance - No Configuration Change - Necessary Data after Shutdown
Issues/Decisions:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00731 Startup all VIL Instances - Configuration Change (Draft) - Default Values after Inconsistencies - General			
In case of data inconsistencies after the startup of a [VIL] with configuration change, the [VIL] shall retrieve the following default values:			
•Current system data (subsidiary or caution signal): subsidiary signal •Current system data (point machine or double key lock at a point): point machine •Release status (single key lock, double key lock): [NADA] •Current number of the registration of registered actions: 0 •Last status of locks: not locked •Last status of indication signs set: indication signs not set •Last status of [SCWS-S]: troop not set •Last status of train movement lock: not set			
Comments: Function according to ELEKTRA 2 baseline 29.20.			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_131] 5.4.0-1 Für den Fall einer Außerbetriebsetzung des VSTW (Außerbetriebsetzung aller VSTW-Partner):
Target Release:	PV_EL_GEN_2.20.0	Scope:	K... System
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test
See also:			
Issues/Decisions:			

EL4-DIL-SSS-REQ-00663 Startup all VIL Instances - No Configuration (Draft) Change - Necessary Data after long-term Shutdown - Line Block			
Concerning "Line Block", after a long-term decommissioning, the [DIL] shall retrieve the values according to chapter [Line Block - Procedure in case of Irregularities].			
Comments: Function according to ELEKTRA 2 baseline 29.20.			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_132] 5.4.0-2 Für den Streckenblock sind die Angaben zu Daten nach einer langfristigen Außerbetriebsetzung...
Target Release:	PV_EL_GEN_2.20.0	Scope:	System
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test
See also:			
Issues/Decisions:			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00664 Startup all VIL Instances - No Configuration Change - Point - Occupation State (Draft)	
For points that are occupied at the time of the recommissioning of a [DIL] after a long-term failure, both legs of the point shall be considered occupied.	
Comments: Function according to ELEKTRA 2 baseline 29.20.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Priority: Mandatory Origin: ==> [EL4 PH] [05_133] 5.4.0-3 Bei Weichen, die zum Zeitpunkt des Rechnerneustarts nach einem langfristigen Ausfall besetzt... Scope: System Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00665 Startup all VIL Instances - No Configuration Change - FLAB Entries (Draft)	
After the recommissioning of a [DIL] after a long-term failure the [FLAB] entries shall be recalculated.	
Comments: Function according to ELEKTRA 2 baseline 29.20.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Priority: Mandatory Origin: ==> [EL4 PH] [05_133] 5.4.0-3 Bei Weichen, die zum Zeitpunkt des Rechnerneustarts nach einem langfristigen Ausfall besetzt... Scope: System Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00666 Startup all VIL Instances - No Configuration Change - WIG Entries (Draft)	
After the recommissioning of a [DIL] after a long-term failure the [WIG] entries shall be recalculated.	
Comments: Function according to ELEKTRA 2 baseline 29.20.	
Category: FUR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Priority: Mandatory Origin: ==> [EL4 PH] [05_134] 5.4.0-4 Die WIG-Einträge sind nach einem Rechnerneustart neu zu berechnen. Scope: System Verification: System test See also:
Issues/Decisions:	

3.2.3.5 RECOMMISSIONING COMMAND AFTER DECOMMISSIONING OF ALL [VIL] INSTANCES

The operator is responsible that the trains are at standstill when applying the *recommissioning command* [WIN].

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00667 Recommissioning Command - Command Filter (Draft)			
After the startup of the [VIL] no operator command shall be possible except for the recommissioning command.			
Comments: Function according to ELEKTRA 2 baseline 29.20.			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_136] 5.5.0-1 Nach dem Hochfahren des VSTW darf außer der Wiederinbetriebnahmebedienung (Hochlaufbedienung...
Target Release:	PV_EL_GEN_2.20.0	Scope:	System
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test
See also:			
Issues/Decisions:			

EL4-DIL-SSS-REQ-00668 Recommissioning Command - Conditions for Request (Draft)			
The [VIL] shall send a request for the recommissioning command in text form only if all necessary data are available and their effects are displayed on the display medium.			
Comments: •Function according to ELEKTRA 2 baseline 29.20. •The request is sent via [X25] to [OMC].			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_136] 5.5.0-1 Nach dem Hochfahren des VSTW darf außer der Wiederinbetriebnahmebedienung (Hochlaufbedienung...
Target Release:	PV_EL_GEN_2.20.0	Scope:	System
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test
See also:			
Issues/Decisions:			

EL4-DIL-SSS-REQ-00669 Recommissioning Command - Restore State (Draft)			
The recommissioning command by the [VIL] shall indicate, if all status data have been restored or not.			
Comments: Function according to ELEKTRA 2 baseline 29.20.			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [05_137] 5.5.0-2 In der Ausgabe des Textes für die Hochlaufbedienung muss unterschieden werden, ob das VSTW ü...
Target Release:	PV_EL_GEN_2.20.0	Scope:	System
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test
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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00670		Recommissioning Command - Documented Action		(Draft)
The recommissioning command shall be a documented action.				
Comments:		Function according to ELEKTRA 2 baseline 29.20.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [05_139] 5.5.0-4 Die Hochlaufbedienung muss eine dokumentationspflichtige Handlung sein.	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Application	Verification:	System test	
		See also:		
Issues/Decisions:				

3.3 D2.2 RAMS

3.3.1 GENERAL

3.3.1.1 ASSUMPTIONS

Assumption 1:

No safety level is required for power supply and maintenance interfaces.

Assumption 2:

No safety level is required for the [DDSP].

Assumption 3:

No safety level is required for all security issues in the sense of safety.

Assumption 4:

Concerning the availability calculation for the manual repair of a disturbance an [MRT] of 0,5h is assumed.

Assumption 5:

Concerning the availability calculation for the manual repair of an error an [MRT] of 1h is assumed.

3.3.1.2 MINIMUM OPERATING LIFE

The minimum operating life is specified in the service agreement and can be guaranteed by measures taken by the supplier, e.g. provision of spare parts/replacement products.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00299		Minimum Operating Life - COTS Components		(Draft)
The minimum operating life of the [COTS] components of the interior system shall be at least 7 years without interruption.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_16] 2.1.0-1 Die Mindestnutzungsdauer der COTS-Komponenten der Innenanlage muss mindestens 7 Jahre betrag...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00300		Minimum Operating Life - Continuous Operation		(Draft)
The supplier shall guarantee the continuous operation of the [DIL] system within the minimum operating life.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_17] 2.1.0-2 Die durchgehende Nutzung des Systems DSTW innerhalb der Mindestnutzungsdauer muss durch den ...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00301		Minimum Operating Life - Extension		(Draft)
he minimum operating life of the [DIL] components of the outdoor system can be extended once by 5 years.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_19] 2.2.0-1 Die Mindestnutzungsdauer der DSTW-Komponenten der Außenanlage kann einmalig um 5 Jahre verlä...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.3.1.3 SPARE PARTS

EL4-DIL-SSS-REQ-00302 Supply for Spare Parts (Draft)	
The supplier of the [DIL] components shall ensure the supply of all spare parts to maintain operation and full functionality for the minimum utilization period of 7 years for the components of the indoor system and 25 years for the components of the outdoor system.	
<i>Comments:</i>	
Category: DPR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_21] 2.3.0-1 Der AN der DSTW-Komponenten muss die Versorgung mit sämtlichen Ersatzteilen zur Aufrechthalt...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
<i>Issues/Decisions:</i>	

3.3.1.4 OEVE/OENORM AND EULYNX

EL4-DIL-SSS-REQ-00303 Fulfillment of Standard EN 50126 (Draft)	
The [DIL] system shall fulfill the requirements according to [OEVE/OENORM EN 50126].	
<i>Comments:</i>	
Category: GCR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_23] 2.4.0-1 Die Anforderungen gemäß [OEVE/OENORM EN 50126] müssen erfüllt werden.
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00304 Fulfillment of Standard EN 50128 (Draft)	
The [DIL] system shall fulfill the requirements according to [OEVE/OENORM EN 50128].	
<i>Comments:</i>	
Category: GCR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_25] 2.5.0-1 Die Anforderungen gemäß [OEVE/OENORM EN 50128] müssen erfüllt werden.
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00305		Fulfillment of Standard EN 50129		(Draft)			
The [DIL] system shall fulfill the requirements according to [OEVE/OENORM EN 50129].							
Comments:							
Category:		GCR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_27] 2.6.0-1 Die Anforderungen gemäß [OEVE/OENORM EN 50129] müssen erfüllt werden.	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00		Verification:		System test	
		DIGITAL INTERLOCKING (DIL)		See also:			
		VIRTUAL INTERLOCKING (VIL)					
		INTEGRATED PLATFORM					
		SIGNALLING (IPS)					
		DIGITAL FIELD ELEMENTS (DFE)					
Issues/Decisions:							

EL4-DIL-SSS-REQ-00306		Fulfillment of EULYNX Standards Eu.Doc.93 and Eu.Doc.16		(Draft)
The [DIL] system shall fulfill the requirements according to [Eu.Doc.93] and [Eu.Doc.16].				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_29] 2.7.0-1 Die Anforderungen gemäß Eu.Doc.93 und Eu.Doc.16 müssen erfüllt werden.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

3.3.2 RELIABILITY

3.3.2.1 DEFINITIONS

Reliability:

Ability to function as required without failure under given conditions and for a given time interval (definition according to [OEVE/OENORM EN 50126]).

Faults/Errors and Disturbances:

Definition and description according to service contract.

Error classes:

•FK1

Error class 1: 'EMERGENCY'

Errors that significantly hinder operations.

•FK2

Error class 2: 'CRITICAL'

Errors that have a significant impact on parts (functional/local) of the operation or serious errors.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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- FK3

Error class 3 'MAJOR'

Other errors or significant errors.

- FK4

Error class 4 'MINOR'

Other errors in the [DIL-CU] or [EIF].

3.3.2.2 DISTURBANCES, ERRORS/FAULTS AND FAILURES

EL4-DIL-SSS-REQ-00316		Reliability - Safe Design		(Draft)			
The [DIL] system with its interfaces shall be designed in such a way that neither faults nor disturbances lead to unsafe conditions.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_35] 3.2.0-1 Das Gesamtsystem DSTW mit seinen Schnittstellen muss derart gestaltet werden, dass weder Feh...	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)		Verification:		System test	
Issues/Decisions:				See also:			

EL4-DIL-SSS-REQ-00317		Reliability - Mathematical Proof		(Draft)	
The supplier shall provide a mathematical proof of the required reliability in accordance with [CENELEC] specifications.					
Comments:					
Category:	DOR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_37] 3.2.0-3 Ein rechnerischer Nachweis für die geforderte Zuverlässigkeit muss vom AN erbracht werden.		
Target Release:	PV_EL_GEN_2.20.0	Scope:	D... System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00318 Reliability - Power Supply Failure - Safe State (Draft)	
In case of a power supply failure, all affected [VIL]s and all associated components of the outdoor system shall enter a safe state.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_39] 3.3.0-1 Bei Ausfall der Stromversorgung müssen alle betroffenen VSTWs mit allen zugeordneten Komponente... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00319 Reliability - Error Reporting - OMC (Draft)	
Disturbances, failures and faults of the [DIL] system shall be reported to the [OMC] system for data recording via interface IF-61 ([X25] over [IP] with [RaSTA]).	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_41] 3.4.0-1 Störungen, Ausfälle und Fehler des Systems DSTW müssen zur Datenregistrierung an das Umsyste... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00320 Reliability - Error Reporting - DDSP via ADM (Draft)	
Disturbances, failures and faults of the [DIL] indoor system and its interfaces shall be reported to the [DDSP] via component [ADM].	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_42] 3.4.0-2 Störungen, Ausfälle und Fehler des Systems DSTW und seiner Schnittstellen müssen an das DDSP... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00321 Reliability - Error Reporting - DDSP via SDI and SSI (Draft)	
Disturbances, failures and faults of the [DIL] outdoor system shall be reported to the [DDSP] via the [EULYNX] interfaces [SDI] and [SSI].	
Comments:	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_42] 3.4.0-2 Störungen, Ausfälle und Fehler des Systems DSTW und seiner Schnittstellen müssen an das DDSP...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00649 Reliability - Resolution of Faults - Responsibility (In negotiation)	
Faults shall be resolved by the maintenance department. Faults shall not require the operator to take any maintenance-related substitute measures or maintenance activities.	
Comments: == OPEN POINT == Process requirement! Does not address system!	
Category: MDR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_44] 3.5.0-1 Fehler dürfen keine instandhaltungstechnischen Ersatzmaßnahmen des Fdl erfordern und er darf...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL)	Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00322 Reliability - Single HW Failure (Draft)	
A single failure of a [HW] component shall not lead to any operational restrictions.	
Comments:	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_46] 3.6.0-1 Ein Tausch ausgefallener/defekter Komponenten muss rückwirkungsfrei auf den Betrieb und die ...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
Issues/Decisions:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00323 Reliability - Failure of a server (Draft)	
A failure of one server shall not affect the operational availability of other servers.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00324 Reliability - Failure of a PDU (Draft)	
A failure of a [PDU] shall not have operational consequences.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_46] 3.6.0-1 Ein Tausch ausgefallener/defekter Komponenten muss rückwirkungsfrei auf den Betrieb und die ... <i>Scope:</i> System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00325 Reliability - Failure of a UPS (Draft)	
A failure of a [UPS] shall not have operational consequences.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_46] 3.6.0-1 Ein Tausch ausgefallener/defekter Komponenten muss rückwirkungsfrei auf den Betrieb und die ... <i>Scope:</i> System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00326 Reliability - Failure of a Connection		(Draft)
A failure of one connection shall not affect the operational availability of the other connections.		
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
<i>Category:</i>	FUR	<i>Priority:</i> Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i>
<i>Target Release:</i>	PV_EL_GEN_2.20.0	<i>Scope:</i> System
	BL_OBB_40.00	
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL)	<i>Verification:</i> Integration test
	VIRTUAL INTERLOCKING (VIL)	<i>See also:</i>
	INTEGRATED PLATFORM	
	SIGNALLING (IPS)	
<i>Issues/Decisions:</i>		

EL4-DIL-SSS-REQ-00327 Reliability - Failure of a User Interface		(Draft)
The failure or malfunction of one user interface shall not affect the operation of another user interface.		
<i>Comments:</i>		
<i>Category:</i>	FUR	<i>Priority:</i> Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i>
<i>Target Release:</i>	PV_EL_GEN_2.20.0	<i>Scope:</i> System
	BL_OBB_40.00	
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL)	<i>Verification:</i> System test
	ELEKTRA Generic System	<i>See also:</i>
	INTEGRATED PLATFORM	
	SIGNALLING (IPS)	
<i>Issues/Decisions:</i>		

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00650		Reliability - Failure of a Global Function		(In negotiation)	
It shall be possible to replace the global functions during normal operation. The system shall be able to handle a failure of these functions and the impact shall be limited to the function being replaced. The impact on operation shall be reduced to a minimum.					
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
		== OPEN POINT ==			
		What are Global Functions ??			
		Anmerkung E. Glaser (02/25):			
		Global Functions“ sind Funktionen die sich auf mehrere VSTWs durchschlagen. z.B. Server, Netzwerk Data Storage, Control Plane, ADM für Diagnose usw			
		Anmerkung G. Eber (03/25):			
		In dieser Form ist dieses Requirement unbrauchbar, weil zu schwammig und daher nicht testbar. Es gibt auch keine entsprechende Kundenanforderungen. Daher würde ich dieses Requirement nicht übernehmen!			
Category:	FUR	Priority:	Mandatory		
Originator:	RAM	Origin:			
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
	BL_OBB_40.00				
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test		
	VIRTUAL INTERLOCKING (VIL)	See also:			
	INTEGRATED PLATFORM				
	SIGNALLING (IPS)				
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00651 Reliability - Duration of Replacement of Global Functions (In negotiation)	
Replacement of global functions during normal operation shall be possible within 15 minutes.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]). == OPEN POINT == What are Global Functions ?? Anmerkung E. Glaser (02/25): Global Functions“ sind Funktionen die sich auf mehrere VSTWs durchschlagen. z.B. Server, Netzwerk Data Storage, Control Plane, ADM für Diagnose usw Anmerkung G. Eber (03/25): In dieser Form ist dieses Requirement unbrauchbar, weil zu schwammig und daher nicht testbar. Es gibt auch keine entsprechende Kundenanforderungen. Daher würde ich dieses Requirement nicht übernehmen!	
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00328 Reliability - Technical single Fault (Draft)	
The occurrence of a technical single fault in the [DIL] system shall not lead to any operational restrictions.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> FUR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> Integration test System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00329 Reliability - Diagnosis of Failure of open Network (Draft)	
The [DIL] shall be able to diagnose failures of the open network (category 2 or 3 according to [EN50159]) and the data terminals with the respective location reference.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> FUR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00330 Reliability - Diagnosis of failure of closed Network (Draft)	
The [DIL] shall be able to diagnose failures of the closed network (category 1 according to [EN50159]) and the data terminals with the respective location reference.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> FUR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

3.3.3 MAINTAINABILITY AND MAINTENANCE

3.3.3.1 DEFINITIONS

Definon "Maintainability":

The ability to remain in, or to be returned to, a functional state, under given application and maintenance conditions (definition according to [OEVE/OENORM EN 50126]).

Definon "Maintenance":

Maintenance is the generic term for inspection, servicing and repair.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.3.3.2 GENERAL

EL4-DIL-SSS-REQ-00331 Maintainability and Maintenance - DDSP Compliance concerning Maintainability (Draft)	
The supplier shall comply with the [DDSP] concept to ensure maintainability.	
<i>Comments:</i>	
<i>Category:</i> DOR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_53] 4.3.0-1 Für die Einhaltung der Instandhaltbarkeit muss das Konzept DDSP vom AN eingehalten werden. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00332 Maintainability and Maintenance - IT Commissioning Compliance (Draft)	
The supplier shall comply with the [IT] commissioning concept to ensure maintainability.	
<i>Comments:</i> This refers to the [LH], part D9 "IT-Inbetriebnahme Konzept", and part D8 "Security Konzept".	
<i>Category:</i> DOR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_54] 4.3.0-2 Für die Einhaltung der Instandhaltbarkeit muss das IT Inbetriebnahmekonzept vom AN eingehalt... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00333		Maintainability and Maintenance - EULYNX Compliance		(Draft)
The supplier shall comply with the relevant documents of the [EULYNX] specification to ensure maintainability.				
Comments:				
Category:	DOR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_55] 4.3.0-3 Für die Einhaltung der Instandhaltbarkeit müssen die relevanten Dokumente der EULYNX-Spezifi...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

3.3.3.3 STORAGE OF SPARE PARTS

EL4-DIL-SSS-REQ-00334		Maintainability and Maintenance - Spare Parts Storage - Concept		(In negotiation)
The supplier shall develop a storage concept for the spare parts as described in the service description.				
Comments: == MISMATCH == Das ist eine Anforderung, aber im LH nur als "Info" markiert.				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_57] 4.4.0-1 Der AN entwickelt ein Konzept der Lagerung für die Ersatzteile, wie in der Leistungsbeschrei...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00335 Maintainability and Maintenance - Spare Parts Storage - Information (Draft)	
The supplier shall provide any information and facilities for spare parts storage.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_58] 4.4.0-2 Allfällige Angaben und Einrichtungen zur Ersatzteillagerung müssen vom AN bereit gestellt we... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00336 Maintainability and Maintenance - Spare Parts Storage - Effort (Draft)	
Basically components shall be developed in such a way that storage is possible with minimal effort.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_58] 4.4.0-2 Allfällige Angaben und Einrichtungen zur Ersatzteillagerung müssen vom AN bereit gestellt we... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00337 Maintainability and Maintenance - Spare Parts Storage - Maintenance (Draft)	
Il spare parts shall be storable without requiring regular maintenance and without being kept under voltage.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_59] 4.4.0-3 Alle Ersatzteile müssen lagerfähig sein, ohne dass sie einer regelmäßigen Instandhaltung unt... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00338		Maintainability and Maintenance - Spare Parts		(Draft)			
Storage - Climate Classes							
Spare parts shall be storable under the environmental conditions in accordance with the climate classes in [03 Lastenheft Allgemeine Anforderungen 1].							
Comments:							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_60] 4.4.0-4 Ersatzteile müssen unter den Umgebungsbedingungen gemäß den Klimaklassen in dem <03 Lastenhe...	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)		Verification:		System test	
				See also:			
Issues/Decisions:							

3.3.3.4 LABELLING

EL4-DIL-SSS-REQ-00339		Maintainability and Maintenance - Labelling - Data		(Draft)
All components and modules of the indoor and outdoor systems, including switch cabinets and switch boxes, shall be clearly labelled in machine-readable form using a [QR] code and shall show at least the following data:				
•Manufacturer •Type •Item number •Device version (hardware and firmware version, etc.) •Serial number •[MAC] addresses				
The customer shall specify the concrete form with the supplier during the specification phase.				
Comments:		Hitachi Rail applies the GS1 data matrix code (2D barcode).		
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_62] 4.5.0-1 Sämtliche Komponenten, Module der Innen- und Außenanlage inkl. Schaltschränke und Schaltkäst...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00340		Maintainability and Maintenance - Labelling - Units		(Draft)
Labelling shall be applied to the smallest replaceable unit, for example: •Object Controller •Power supply units •Light point •Wheel sensor •Lightning protection •Signal box •Separately mountable parts by the supplier •[FES] •Server				
The customer shall specify the details with the supplier during the specification phase.				
Comments:				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_63] 4.5.0-2 Die Kennzeichnung erfolgt auf die kleinste Tauschbare Einheit, Liste beispielhaft: • Object ...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00341		Maintainability and Maintenance - Labelling - QR-Code for Outdoor System		(Draft)
The [QR] code for the outdoor system shall be weatherproof and installed appropriately.				
Comments:				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_64] 4.5.0-3 Der QR-Code für Außenanlage muss wetterfest/witterungsfest und entsprechend montiert sein.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00342 Maintainability and Maintenance - Labelling - QR-Code - Format (Draft)	
The supplier shall comply with the [QR] code standard and data format.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_65] 4.5.0-4 Der AN muss den QR-Code Standard bzw. Datenformat einhalten. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00343 Maintainability and Maintenance - Labelling - Serial Number (Draft)	
The serial number of the components in the spare parts inventory shall enable clear identification of the components throughout their entire operating life.	
<i>Comments:</i> The serial numbers of the components allow for transparent tracking of components throughout their entire operating life (e.g. in [DDSP], [AVS], [SAP]).	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_66] 4.5.0-5 Die Seriennummer der Komponenten der Ersatzteilkhaltung muss eine eindeutige Identifikation d... ==> [EL4 PH] [02_67] 4.5.0-6 Die Seriennummer der Komponenten dient der transparenten Verfolgung der Komponenten über die... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

3.3.3.5 MAINTENANCE DOCUMENTATION

Maintenance is carried out in accordance with the service description and service level agreement.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00345		Maintainability and Maintenance - Documentation - Useability		(Draft)
The supplier shall deliver the maintenance documentation in a form that it enables the customer to perform the inspection, maintenance and repair (at least component replacement, troubleshooting, resetting, restarting, etc.) of the supplied system.				
Comments:				
Category:	DOR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_70] 4.6.0-2 Die Instandhaltungsdokumentation muss die Inspektion, Wartung und Instandsetzung (mindestens...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00346		Maintainability and Maintenance - Documentation - Generic Form		(Draft)
The supplier shall prepare the maintenance documentation generically, so that it is applicable to all parts of the supplied system.				
Comments:		<u>Note:</u> The preparation of a site specific maintenance documentation by the supplier is not planned.		
Category:	DOR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_71] 4.6.0-3 Die Instandhaltungsdokumentation muss generisch erstellt und für sämtliche vom AN gelieferte... ==> [EL4 PH] [02_72] 4.6.0-4 Es ist nicht vorgesehen, dass der AN eine anlagenspezifische Instandhaltungsdokumentation li...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00347 Maintainability and Maintenance - Documentation - Electronic Form (Draft)	
The supplier shall provide the maintenance documentation in printed and electronic form and make it available for the [DDSP].	
Comments: Note: The documentation is transferred to the customer.	
Category: DOR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Priority: Mandatory Origin: ==> [EL4 PH] [02_73] 4.6.0-5 Die Instandhaltungsdokumentation muss gedruckt, elektronisch und für das DDSP zur Verfügung ... Scope: System Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00348 Maintainability and Maintenance - Documentation - File Format (Draft)	
The file format for the maintenance documentation in [DDSP] shall be agreed between the customer and the supplier during the specification phase.	
Comments:	
Category: DOR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ELEKTRA Generic System INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Priority: Mandatory Origin: ==> [EL4 PH] [02_74] 4.6.0-6 Das Dateiformat für die Instandhaltungsdokumentation im DDSP ist während der Pflichtenheftph... Scope: System Verification: System test See also:
Issues/Decisions:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.3.3.6 MAINTENANCE PROCESS

EL4-DIL-SSS-REQ-00349 Maintainability and Maintenance - Process - Implementation and Practice (Draft)	
The supplier shall implement a maintenance process and practice it with the customer.	
Comments: == OPEN POINT == This is a process requirement!	
Category: MDR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL)	Priority: Mandatory Origin: ==> [EL4 PH] [02_76] 4.7.0-1 Es muss ein Instandhaltungsprozess zwischen AN und AG implementiert und gelebt werden. Systeme... Scope: System Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00817 Maintainability and Maintenance - Process - DDSP Compliance concerning Maintenance (Draft)	
The supplier shall ensure that the supplied systems, subsystems and components comply with the maintenance process, taking into account the [DDSP] concept.	
Comments: == OPEN POINT == This is a process requirement!	
Category: MDR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL)	Priority: Mandatory Origin: ==> [EL4 PH] [02_76] 4.7.0-1 Es muss ein Instandhaltungsprozess zwischen AN und AG implementiert und gelebt werden. Systeme... Scope: System Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00350 Maintainability and Maintenance - Process - Access (In negotiation)	
Maintenance staff shall be able to access the relevant service documentation (locally or from the [DDSP]) by using the [QR] code on the individual elements (e.g. [OC]).	
Comments: == MISMATCH == Das ist eine Anforderung, aber im LH nur als "Info" markiert.	
Category: MDR Originator: Customer Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Priority: Mandatory Origin: ==> [EL4 PH] [02_77] 4.7.0-2 Mittels QR-Code an den einzelnen Elementen (z.B. OC) soll der Instandhalter die dazugehörige... Scope: System Verification: System test See also:
Issues/Decisions:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00351 Maintainability and Maintenance - Process - Manual Earth Fault Measurements (Draft)	
Manual earth fault measurements shall not be permitted during operation.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_90] 4.9.0-4 Im Betrieb sind keine manuellen Erdschlussmessungen zugelassen. <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

3.3.3.7 COMPONENT EXCHANGE

EL4-DIL-SSS-REQ-00352 Maintainability and Maintenance - Component Exchange - Restrictions (Draft)	
It shall not be possible to reconfigure modules or change the data preparation of modules if they have been replaced by maintenance personnel due to faults or disturbances.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_79] 4.8.0-1 Eine Neukonfiguration oder Umprojektierung von Modulen darf bei fehler- und störungsbedingte... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00353 Maintainability and Maintenance - Component Exchange - OC Replacement - Restrictions (Draft)	
The [OC]s shall be designed in that way that it is not possible to replace them with the wrong spare part.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_80] 4.8.0-2 Object Controller müssen so konstruiert sein, dass der Tausch mit einem falschen Ersatzteil ... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00354 Maintainability and Maintenance - Component Exchange - OC Replacement - Impact (Draft)	
The replacement of an [OC] shall not result in any operational effects or disturbances other than the replacement of this [OC].	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_81] 4.8.0-3 Der Tausch eines Object Controllers darf über den Tausch dieses Object Controllers hinaus zu... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00355 Maintainability and Maintenance - Component Exchange - Approved Tools (Draft)	
The assembly, disassembly and repair of components shall be carried out with tools that have been approved by the customer.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_82] 4.8.0-4 Die Montage, Demontage und Instandsetzung von Komponenten muss mit Werkzeugen ausgeführt wer... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00356 Maintainability and Maintenance - Component Exchange - Standard Tools (Draft)	
No specialized tool (designed to work only with the component being replaced) shall be required to replace any component.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_83] 4.8.0-5 Für den Tausch von Komponenten darf kein Spezialwerkzeug (welches nur für den Einsatzbereich... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00357 Maintainability and Maintenance - Component Exchange - Manual corrective Maintenance Actions (Draft)	
The manual corrective maintenance actions of a [HW] component (incl. router, switches, control plane, computing server, storage engine) like exchange, cabling, etc. up to the successful power-on shall be done within 45 minutes for a required [MRT] of 1 hour.	
Comments:	
Category: MDR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_84] 4.8.0-6 Für den Tausch von Komponenten muss im Zuge der Instandhaltung nachweislich der geforderte M... Scope: System
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
Issues/Decisions:	

EL4-DIL-SSS-REQ-00358 Maintainability and Maintenance - Component Exchange - Sufficent Data to DDSP (Draft)	
The [DIL-CU] shall deliver all the necessary data to the [DDSP] so that all required information for maintenance activities concerning component exchange can be derived from the [DDSP].	
Comments: •The customer provides as software tool a data diagnosis service platfrom [DDSP]. •Avoiding additional apps/applications/software requires the integration of third components into the [DDSP]. == OPEN POINT == Specification of the interface between [DIL] and [DDSP]	
Category: MDR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_85] 4.8.0-7 Als Softwarewerkzeug für die Instandhaltung stellt der AG eine Daten-Diagnose-Service-Plattf... ==> [EL4 PH] [02_86] 4.8.0-8 Für die Instandhaltung dürfen zur DDSP keine zusätzlichen Apps/Applikationen/Software benöti... Scope: System
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
Issues/Decisions:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00359		Maintainability and Maintenance - Inspection- and Maintenance Freeness		(Draft)	
The following supplier-provided components shall be inspection- and maintenance-free: •[DC] components •Field elements (including their stations) and [OC]s <u>Exceptions:</u> •Cleaning of the signal optics of the light signal subsystem •Adjusting of the axle counter of the part system [TDS]. If this cannot be achieved, a minimum maintenance interval of 6 months shall be allowed.					
<i>Comments:</i>		<u>Note:</u>			
		The filter mats and fans of network components are cleaned or replaced regularly.			
<i>Category:</i> MDR <i>Originator:</i> Customer		<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_88] 4.9.0-1 Alle Komponenten des Rechenzentrums, die vom AN geliefert werden, müssen inspektions- und wa... ==> [EL4 PH] [02_89] 4.9.0-2 Alle Feldelemente müssen inspektionsfrei sein und dürfen keine regelmäßige Wartungsmaßnahmen... ==> [EL4 PH] [02_258] 4.9.0-3 Feldelementstationen und Object Controller müssen wartungsfrei sein.			
<i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)		<i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>			
<i>Issues/Decisions:</i>					

EL4-DIL-SSS-REQ-00360		Maintainability and Maintenance - Freeness of preventive Maintenance		(Draft)	
The [DIL] shall be free of preventive maintenance activities. If this is not technologically feasible for components or if the performance or service life is increased as a result, a maximum of a half-year period maintenance interval shall be granted.					
<i>Comments:</i>		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory		
<i>Originator:</i>	RAM	<i>Origin:</i>			
<i>Target Release:</i>	PV_EL_GEN_2.20.0	<i>Scope:</i>	System		
	BL_OBB_40.00				
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL)	<i>Verification:</i>	System test		
	INTEGRATED PLATFORM	<i>See also:</i>			
	SIGNALLING (IPS)				
	ADMINISTRATION, DIAGNOSTIC				
	AND MAINTENANCE-SERVICES				
	(ADM)				
	DIGITAL FIELD ELEMENTS (DFE)				
<i>Issues/Decisions:</i>					

Hitachi Rail – Internal			
Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00361 Maintainability and Maintenance - Easy Accessibility		(Draft)
Servers and cables shall be easily accessible for maintenance. Therefore, the servers shall be able to be pulled out fully during operation and cable management shall be done with cable management arms.		
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
<i>Category:</i> MDR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> Integration test <i>See also:</i>	
<i>Issues/Decisions:</i>		

3.3.3.9 EMPLOYEE PROTECTION

EL4-DIL-SSS-REQ-00566 Maintainability and Maintenance - Employee Protection		(Draft)
Components, subsystems and systems supplied by the supplier shall be designed in such a way that the employee protection is ensured when the customer carries out maintenance works.		
<i>Comments:</i> The employee protection for the customer's employees can only be ensured by the customer himself.		
<i>Category:</i> GCR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_92] 4.10.0-1 Komponenten, Teilsysteme und Systeme, welche durch den AN geliefert werden, müssen so konzi... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>	
<i>Issues/Decisions:</i>		

3.3.4 AVAILABILITY

3.3.4.1 DEFINITIONS

Availability:

The ability to perform a required function at a given time or during a given time interval under given conditions, provided that the necessary external aids are provided (definition according to [OEVE/OENORM EN 50126]).

Automatically rectified:

In the context of a safe computer unit means the following:

After a disturbance has been detected, a safe computer unit enters a safe state and performs a self-test. After a successful self-test, the disturbed module is automatically recommissioned.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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This also applies to the other availability requirements in this chapter.

3.3.4.2 AVAILABILITY CLASSES

Depending on the required availability values, the following availability classes are defined:

- Class 1: Availability is 0.99999
- Class 2: Availability is 0.99995
- Class 3: Availability is 0.9999

The availability values were determined in accordance with [OEVE/OENORM EN 50126] in a top-down approach as follows:

$$A = [MTTF]/([MTTF]+[MTTR])$$

3.3.4.3 DIL-CU AVAILABILITY

3.3.4.3.1 GENERAL

The [20 Lastenheft Rechenzentrum 1] is a unit of all [DIL-CU] ([VIL] instances including their hardware), network technology, air conditioning, additional applications ([DDSP]), [UPS] and its physical location. The data centre part system is distributed across at least two locations and is specified by the customer.

MLS Allocation: DIGITAL INTERLOCKING (DIL)

VIRTUAL INTERLOCKING (VIL)

INTEGRATED PLATFORM SIGNALLING (IPS)

EL4-DIL-SSS-REQ-00567		Availability - Combined SW Package		(Draft)			
In order to be able to comply with the availability of the [DIL-CU], the supplier shall provide the operating system and the operational software.							
Comments:							
Category:		GCR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_101] 5.3.0-2 Um die Verfügbarkeiten der DSTW-ZE einhalten zu können, muss der AN das Betriebssystem und d...	
Target Release:		PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope:		System	
Allocation:		DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)		Verification:		System test	
				See also:			
Issues/Decisions:							

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00568		Availability - Responsibility		(Draft)	
The supplier shall take responsibility for all components/systems of the [DIL-CU] (SW, patch management, operating system, operational software, security, etc.) that are involved and that are supplied or provided by the supplier.					
Comments:					
Category: GCR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [02_102] 5.3.0-3 Der AN muss für alle beteiligten, von ihm gelieferten bzw. ihm beigestellten, Komponenten/Sy...			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope: System			
Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)		Verification: System test			
		See also:			
Issues/Decisions:					

EL4-DIL-SSS-REQ-00406		Availability - DIL-CU - Overall		(Draft)	
The [DIL-CU] shall have a class 1 overall availability of at least 0,99999.					
Comments:					
Category: DPR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [02_104] 5.3.0-5 Die Verfügbarkeit der DSTW-ZE muss mindestens 0,99999 betragen. ==> [EL4 PH] [02_106] 5.3.0-7 Jede DSTW-ZE wird der Verfügbarkeitsklasse 1 zugeordnet.			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope: System			
Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)		Verification: System test See also:			
Issues/Decisions:					

EL4-DIL-SSS-REQ-00407		Availability - DIL-CU - Disturbances - Manual Rectification		(Draft)	
The [DIL-CU] shall have a class 1 availability of at least 0,99998 regarding to disturbances that need to be rectified manually.					
Comments:					
Category: DPR		Priority: Mandatory		Origin: ==> [EL4 PH] [02_107] 5.3.0-8 Die Verfügbarkeit der DSTW-ZE in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 1) ...	
Originator: Customer		Origin:			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope: System			
Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)		Verification: System test See also:			
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00408 Availability - DIL-CU - Errors without operational Impact (Draft)			
The [DIL-CU] shall have a class 1 availability of at least 0,997154 regarding to errors that have no operational impact.			
Comments:			
Category:	DPR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_108] 5.3.0-9 Die Verfügbarkeit der DSTW-ZE in Bezug auf Fehler, welche keine betrieblichen Auswirkungen h...
Target Release:	PV_EL_GEN_2.20.0	Scope:	System
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)	Verification:	System test
See also:			
Issues/Decisions:			

EL4-DIL-SSS-REQ-00409 Availability - DIL-CU - Disturbances - Automatic Rectification - 10 Minutes (Draft)			
The [DIL-CU] shall have a class 1 availability of at least 0,99996 regarding to disturbances that are automatically rectified by the [DIL-CU] within 10 minutes.			
Comments: == OPEN POINT == Shouldn't the value be 0,99999 ??			
Category:	DPR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_109] 5.3.0-10 Die Verfügbarkeit der DSTW-ZE in Bezug auf Störungen, welche durch die DSTW-ZE innerhalb vo...
Target Release:	PV_EL_GEN_2.20.0	Scope:	System
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)	Verification:	System test
See also:			
Issues/Decisions:			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00410		Availability - DIL-CU - Disturbances - Automatic Rectification - 2 Minutes		(Draft)
The [DIL-CU] shall have a class 1 availability of at least 0,99996 regarding to disturbances that are automatically rectified by the [DIL-CU] within 2 minutes.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_110] 5.3.0-11 Die Verfügbarkeit der DSTW-ZE in Bezug auf Störungen, welche durch die DSTW-ZE innerhalb vo...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS)	Verification:	System test	
		See also:		
Issues/Decisions:				

3.3.4.3.2 VIL AVAILABILITY

3.3.4.3.2.1 GENERAL

EL4-DIL-SSS-REQ-00587		Availability - VIL - Overall		(Draft)
The [VIL] shall have a class 2 overall availability of at least 0,99995.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_113] 5.4.0-1 Die Verfügbarkeit des VSTW muss mindestens 0,99995 betragen. ==> [EL4 PH] [02_115] 5.4.0-3 Jedes VSTW wird der Verfügbarkeitsklasse 2 zugeordnet.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00569		Availability - VIL - Disturbances - Manual Rectification		(Draft)
The [VIL] shall have a class 2 availability of at least 0,999979 regarding to disturbances that need to be rectified manually.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_116] 5.4.0-4 Die Verfügbarkeit des VSTW in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 2) mus...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	System test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00570		Availability - VIL - Errors without operational Impact		(Draft)
The [VIL] shall have a class 2 availability of at least 0,997154 regarding to errors that have no operational impact.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_117] 5.4.0-5 Die Verfügbarkeit des VSTW in Bezug auf Fehler, welche keine betrieblichen Auswirkungen habe...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00571		Availability - VIL - Disturbances - Automatic Rectification - 10 Minutes		(Draft)
The [VIL] shall have a class 2 availability of at least 0,99999 regarding to disturbances that are automatically rectified by the [VIL] within 10 minutes.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_118] 5.4.0-6 Die Verfügbarkeit des VSTW in Bezug auf Störungen, welche durch das VSTW innerhalb von 10 Mi...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00572		Availability - VIL - Disturbances - Automatic Rectification - 2 Minutes		(Draft)
The [VIL] shall have a class 1 availability of at least 0,99996 regarding to disturbances that are automatically rectified by the [VIL] within 2 minutes.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_119] 5.4.0-7 Die Verfügbarkeit des VSTW in Bezug auf Störungen, welche durch das VSTW innerhalb von 2 Min...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00573		Availability - VIL - Minor SW Updates		(Draft)	
Minor updates of the [VIL] [SW] (= Patches) shall be carried out without operational restrictions.					
Comments:		•Examples for [SW] Patches: •Security update •Update of application properties (e.g. buffer size, memory size) •See also concept document [Digitales Stellwerk - Technische Konzepte] chapter 9.			
Category:	MCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_120] 5.4.0-8 Erweiterungen und/oder Änderungen der VSTW-Software (Software Updates/Upgrades, Projektierung... ==> [EL4 PH] [02_152] 5.9.0-1 Ein planmäßiges Servicesoftwareupgrade bzw. ein planmäßiger Hardwaretausch muss rückwirkungs...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	System test		
Issues/Decisions:		See also:			

EL4-DIL-SSS-REQ-00818		Availability - VIL - Major SW Updates		(Draft)	
Major updates of the [VIL] [SW], especially with safety impact, (= Upgrades) shall be carried out during operational breaks according to the [DDSP] Use Case SysF4 "System daten aktivieren".					
Comments:		•Examples for [SW] Upgrades: •Enhancement or correction of the application [SW] •Update of the data preparation •See also concept document [Digitales Stellwerk - Technische Konzepte] chapter 9.			
Category:	MCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_120] 5.4.0-8 Erweiterungen und/oder Änderungen der VSTW-Software (Software Updates/Upgrades, Projektierung... ==> [EL4 PH] [02_152] 5.9.0-1 Ein planmäßiges Servicesoftwareupgrade bzw. ein planmäßiger Hardwaretausch muss rückwirkungs...		
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System		
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	System test		
Issues/Decisions:		See also:			

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00574 Availability - VIL - Operational Downtime (Draft)	
The operational downtime shall not be longer than the startup time of the affected component, as well as the time for the re-integration into the overall system.	
<i>Comments:</i>	
<i>Category:</i> MCR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_120] 5.4.0-8 Erweiterungen und/oder Änderungen der VSTW-Software (Software Updates/Upgrades, Projektierung... System <i>Scope:</i> <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

3.3.4.3.2.2 AUTOMATIC TRAIN NUMBER UPDATE

EL4-DIL-SSS-REQ-00362 Availability - Automatic Train Number Update - Overall (Draft)	
The automatic train number update shall have a class 4 overall availability of 99,99 %.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> Analysis Performance test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00363 Availability - Automatic Train Number Update - Distribution - MTTR 6h (Draft)	
The distribution of the availability of automatic train number update shall have a class 4 availability of at least 99,99429 % regarding to a [MTTR] of 6 hours.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> Analysis Performance test <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-DIL-SSS-REQ-00364				Availability - Automatic Train Number Update - Distribution - MTTR 10min		(Draft)	
The distribution of the availability of automatic train number update shall have a class 4 availability of at least 99,99619 % regarding to a [MTTR] of 10 minutes.							
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).					
Category:	DPR	Priority:	Mandatory				
Originator:	RAM	Origin:					
Target Release:	PV_EL_GEN_2.20.0	Scope:	System				
	BL_OBB_40.00						
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	Analysis				
	VIRTUAL INTERLOCKING (VIL)		Performance test				
See also:							
Issues/Decisions:							

EL4-DIL-SSS-REQ-00365				Availability - Automatic Train Number Update - Distribution - MTTR 2min		(Draft)	
The distribution of the availability of automatic train number update shall have a class 4 availability of at least 99,99239 % regarding to a [MTTR] of 2 minutes.							
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).					
Category:	DPR	Priority:	Mandatory				
Originator:	RAM	Origin:					
Target Release:	PV_EL_GEN_2.20.0	Scope:	System				
	BL_OBB_40.00						
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	Analysis				
	VIRTUAL INTERLOCKING (VIL)		Performance test				
See also:							
Issues/Decisions:							

3.3.4.3.2.3 AUTOMATIC FUNCTION

EL4-DIL-SSS-REQ-00366		Availability - Automatic Function - Overall		(Draft)	
The automatic function shall have a class 4 overall availability of 99,99 %.					
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
Category:	DPR	Priority:	Mandatory		
Originator:	RAM	Origin:			
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
	BL_OBB_40.00				
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	Analysis		
	VIRTUAL INTERLOCKING (VIL)		Performance test		
See also:					
Issues/Decisions:					

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00367 Availability - Automatic Function - Distribution - (Draft) MTTR 6h			
The distribution of the availability of the automatic function shall have a class 4 availability of at least 99,99429 % regarding to a [MTTR] of 6 hours.			
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
Category:	DPR	Priority:	Mandatory
Originator:	RAM	Origin:	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	Analysis Performance test
Issues/Decisions:		See also:	

EL4-DIL-SSS-REQ-00368 Availability - Automatic Function - Distribution - (Draft) MTTR 10min			
The distribution of the availability of the automatic function shall have a class 4 availability of at least 99,99619 % regarding to a [MTTR] of 10 minutes.			
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
Category:	DPR	Priority:	Mandatory
Originator:	RAM	Origin:	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	Analysis Performance test
Issues/Decisions:		See also:	

EL4-DIL-SSS-REQ-00369 Availability - Automatic Function - Distribution - (Draft) MTTR 2min			
The distribution of the availability of the automatic function shall have a class 4 availability of at least 99,99239 % regarding to a [MTTR] of 2 minutes.			
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
Category:	DPR	Priority:	Mandatory
Originator:	RAM	Origin:	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	Analysis Performance test
Issues/Decisions:		See also:	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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3.3.4.3.2.4 VIL WORKPLACE

EL4-DIL-SSS-REQ-00374 Availability - VIL Workplace - Overall (Draft)	
A [VIL] workplace related to the [VIL] shall have a class 5 overall availability of 99,98 %.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> Analysis Performance test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00375 Availability - VIL Workplace - Distribution - MTTR 6h (Draft)	
The distribution of the availability of a [VIL] workplace for the [VIL] shall have a class 5 availability of at least 99,98858 % regarding to a [MTTR] of 6 hours.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> Analysis Performance test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00376 Availability - VIL Workplace - Distribution - MTTR 10min (Draft)	
The distribution of the availability of a [VIL] workplace for the [VIL] shall have a class 5 availability of at least 99,99239 % regarding to a [MTTR] of 10 minutes.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> Analysis Performance test <i>See also:</i>
<i>Issues/Decisions:</i>	

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00377		Availability - VIL Workplace - Distribution - MTTR 2min		(Draft)	
The distribution of the availbility of a [VIL] workplace for the [VIL] shall have a class 5 availability of at least 99,98478 % regarding to a [MTTR] of 2 minutes.					
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
Category:	DPR	Priority:	Mandatory		
Originator:	RAM	Origin:			
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
	BL_OBB_40.00				
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	Analysis		
	VIRTUAL INTERLOCKING (VIL)		Performance test		
Issues/Decisions:		See also:			

3.3.4.4 SiDiNet AVAILABILITY

The [SiDiNet] is realised by the customer. The requirements are specified in [22 Network specifications 1] and do not need to be observed by the supplier.

3.3.4.5 FES AVAILABILITY

EL4-DIL-SSS-REQ-00575		Availability - FES - Overall		(Draft)	
The [FES] shall have a class 2 overall availability of at least 0.99995 (without consideration of the availability of the [OC]s).					
Comments:					
Category: DPR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [02_124] 5.6.0-1 Die Verfügbarkeit des FES muss mindestens 0,99995 betragen. Hierbei muss die Verfügbarkeit d...			
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00		Scope: ==> [EL4 PH] [02_126] 5.6.0-3 Die FES (Feldelementstation) wird der Verfügbarkeitsklasse 2 zugeordnet. System			
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)		Verification: Analysis Performance test			
		See also:			
Issues/Decisions:					

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00577		Availability - FES - Disturbances - Manual Rectification		(Draft)
The [FES] shall have a class 2 availability of at least 0.999979 regarding to disturbances that need to be rectified manually.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_127] 5.6.0-4 Die Verfügbarkeit der FES in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 3) muss...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	Analysis Performance test	
Issues/Decisions:		See also:		

EL4-DIL-SSS-REQ-00578		Availability - FES - Errors without operational Impact		(Draft)
The [FES] shall have a class 2 availability of at least 0.997154 regarding to errors that have no operational impact.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_128] 5.6.0-5 Die Verfügbarkeit der FES in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	Analysis Performance test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00579		Availability - FES - Disturbances - Automatic Rectification - 10min		(Draft)
The [FES] shall have a class 2 availability of at least 0,99999 regarding to disturbances that are automatically rectified by the [FES] within 10 minutes.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_129] 5.6.0-6 Die Verfügbarkeit der FES in Bezug auf Störungen, welche in der FES innerhalb von 10 Minuten...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	Analysis Performance test	
Issues/Decisions:		See also:		

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EL4-DIL-SSS-REQ-00580		Availability - FES - Disturbances - Automatic Rectification - 2min		(Draft)
The [FES] shall have a class 2 availability of at least 0,99996 regarding to disturbances that are automatically rectified by the [FES] within 2 minutes.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_130] 5.6.0-7 Die Verfügbarkeit der FES in Bezug auf Störungen, welche in der FES innerhalb von 2 Minuten ...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	Analysis Performance test	
		See also:		
Issues/Decisions:				

3.3.4.6 OC AVAILABILITY

EL4-DIL-SSS-REQ-00581		Availability - OC - Overall		(Draft)
The [OC]s, the block converters and the [LCSI] converters shall have a class 3 overall availability of at least 0.9999.				
Comments: [OC] refers to [LS], [P], [TDS] and [Gen-IO]				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_132] 5.7.0-1 Die Verfügbarkeit der OC (LS, P, TDS, Gen IO), Blockkonverter und EKSA-Umsetzer muss mindest...	
			==> [EL4 PH] [02_134] 5.7.0-3 OC, Blockkonverter und EKSA-Umsetzer werden der Verfügbarkeitsklasse 3 zugeordnet.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	Analysis	
	DIGITAL INTERLOCKING (DIL)		Performance test	
	DIGITAL FIELD ELEMENTS (DFE)			
	OC-P	See also:		
	OC-IO			
	OC-LS			
	OC-TDS			
	IFC-LCSI			
	IFC-Block			
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00582		Availability - OC - Disturbances - Manual Rectification		(Draft)
The [OC]s, the block converters and the [LCSI] converters shall have a class 3 availability of at least 0.999957 regarding to disturbances that need to be rectified manually.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_135] 5.7.0-4 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf manuell zu behebbende...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-P OC-IO OC-LS OC-TDS IFC-LCSI IFC-Block	Verification:	Analysis Performance test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00583		Availability - OC - Errors without operational Impact		(Draft)
The [OC]s, the block converters and the [LCSI] converters shall have a class 3 availability of at least 0,997154 regarding to errors that have no operational impact.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_136] 5.7.0-5 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Fehler, welche keine...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-P OC-IO OC-LS OC-TDS IFC-LCSI IFC-Block	Verification:	Analysis Performance test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00584		Availability - OC - Disturbances - Automatic Rectification - 10min		(Draft)
The [OC]s, the block converters and the [LCSI] converters shall have a class 2 availability of at least 0,99999 regarding to disturbances that are automatically rectified by the [OC]s, the block converters and the [LCSI] converters within 10 minutes.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_137] 5.7.0-6 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Störungen, welche vo...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-P OC-IO OC-LS OC-TDS IFC-LCSI IFC-Block	Verification:	Analysis Performance test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00585		Availability - OC - Disturbances - Automatic Rectification - 2min		(Draft)
The [OC]s, the block converters and the [LCSI] converters shall have a class 2 availability of at least 0,99996 regarding to disturbances that are automatically rectified by the [OC]s, the block converters and the [LCSI] converters within 2 minutes.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_138] 5.7.0-7 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Störungen, welche vo...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-P OC-IO OC-LS OC-TDS IFC-LCSI IFC-Block	Verification:	Analysis Performance test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00586		Availability - OC - Disturbances - SW Changes		(Draft)
If necessary, enhancements and changes of the software of the [OC]s, the block converters and the [LCSI] converters shall be carried out during operational breaks, to avoid a restriction of operation below the permitted availability.				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_139] 5.7.0-8 Erweiterungen und/oder Änderungen der Software von OC, Blockkonverter und EKSA-Umsetzer (Sof...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-P OC-IO OC-LS OC-TDS IFC-LCSI IFC-Block	Verification:	Analysis Performance test	
		See also:		
Issues/Decisions:				

3.3.4.7 LRU AVAILABILITY

EL4-DIL-SSS-REQ-00313		Availability - LRU - No Maintenance Impact		(Draft)
A corrective maintenance action of a [LRU] shall not affect the operational availability of other [LRU]s.				
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category:	DPR	Priority:	Mandatory	
Originator:	RAM	Origin:		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
	BL_OBB_40.00			
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	Integration test	
	DIGITAL FIELD ELEMENTS (DFE)	See also:		
Issues/Decisions:				

3.3.4.8 MONITORING SYSTEM [FRACAS]

The customer provides the supplier with the necessary information from the [MDM]/[DDSP] for the availability verification.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00671		Availability - FRACAS - Compliance with Standards		(Draft)
The supplier shall provide the customer with a [FRACAS] system in accordance with [CENELEC] standard [EN50126-1] for all systems and subsystems to be delivered.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_141] 5.8.0-1 Der AN muss ein FRACAS-System gemäß CENELEC-Norm EN50126-1 dem AG für alle zu liefernden Sys...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00672		Availability - FRACAS - Proof		(Draft)	
The supplier shall prove by means of [FRACAS] that the required availability values ([DC], [VIL], [FES], [OC]) have been met within 5 years of the first [VIL] being put into operation.					
Comments:					
Category:	GCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_142] 5.8.0-2 Der AN muss mittels FRACAS die Einhaltung der geforderten Verfügbarkeitswerte (Rechenzentrum...		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
Allocation:	BL_OBB_40.00	Verification:	System test		
	DIGITAL INTERLOCKING (DIL)	See also:			
	VIRTUAL INTERLOCKING (VIL)				
	INTEGRATED PLATFORM				
	SIGNALLING (IPS)				
	ADMINISTRATION, DIAGNOSTIC				
	AND MAINTENANCE-SERVICES				
	(ADM)				
	DIGITAL FIELD ELEMENTS (DFE)				
Issues/Decisions:					

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00673		Availability - FRACAS - Monitoring and Verifying		(Draft)
The [FRACAS] system shall be used to monitor the availability of each individual [DIL] component ([VIL], [FES], [OC]) and to verify compliance with the availability requirements.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_143] 5.8.0-3 Mittels dem FRACAS-System muss die Verfügbarkeit jeder einzelnen DSTW-Komponenten (VSTW, Fel...	
			==> [EL4 PH] [02_147] 5.8.0-7 Der AN muss die Verfügbarkeit der einzelnen, gelieferten Systeme überwachen und nachweisen.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	ADMINISTRATION, DIAGNOSTIC			
	AND MAINTENANCE-SERVICES			
	(ADM)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00674		Availability - FRACAS - Regular Reports		(Draft)
The [FRACAS] system shall provide the customer with regular reports.				
Comments:				
Category:	DOR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_144] 5.8.0-4 Das FRACAS-System muss dem AG regelmäßige Reports bereitstellen.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	ADMINISTRATION, DIAGNOSTIC			
	AND MAINTENANCE-SERVICES			
	(ADM)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00675		Availability - FRACAS - Display of Information		(Draft)
In reporting, for example, information such as the availability of classes 1 to 3 shall be displayed. In doing so, the information on availability down to the module level (module -> see definition [DIL]) shall be displayed/shown with the information on the reason for failure.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_145] 5.8.0-5 Im Berichtswesen sollen beispielsweise Angaben wie Verfügbarkeit der Klassen 1 bis 3 angezei...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00676		Availability - FRACAS - SLAs		(Draft)
The required availability shall be maintained in line with the availability of OBB's standard fault clearance processes:				
•[SLA] 5 minutes for [VIL]				
•[SLA] 30 minutes for [FES] and [OC].				
Comments:				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_146] 5.8.0-6 Die geforderte Verfügbarkeit ist entsprechend der Verfügbarkeit der Standard ÖBB-Entstörproz...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	INTEGRATED PLATFORM			
	SIGNALLING (IPS)			
	ADMINISTRATION, DIAGNOSTIC			
	AND MAINTENANCE-SERVICES			
	(ADM)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00678		Availability - FRACAS - Monitoring, Documentation and Analysis		(Draft)
The system integrator (supplier of the [VIL]) shall monitor, document and analyse the availability of the overall system, including all subsystems, and communicate the results to the customer.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_149] 5.8.0-9 Der Systemintegrator (Lieferant des VSTW) muss die Verfügbarkeit des Gesamtsystems inklusive...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00679		Availability - FRACAS - Coordination due to Non-Compliance		(In negotiation)
The system integrator (supplier of the [VIL]) shall be responsible for any necessary coordination due to non-compliance with availability.				
Comments: == OPEN POINT == This is a process requirement				
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_150] 5.8.0-10 Die Verantwortung für jegliche notwendige Abstimmung, Aufgrund von nicht einhalten der Verf...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00680		Availability - FRACAS - Cooperation due to Non-Compliance		(In negotiation)
The supplier of the subsystems shall have a constructive obligation to cooperate in the coordination measures mentioned in [EL4-DIL-SSS-REQ-00679].				
Comments: == OPEN POINT == This is a process requirement				
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_150] 5.8.0-10 Die Verantwortung für jegliche notwendige Abstimmung, Aufgrund von nicht einhalten der Verf...	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System	
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00681		Availability - FRACAS - Responsibility		(In negotiation)
The supplier shall have the responsibility for the availability of the system delivered.				
Comments: == OPEN POINT == This is a process requirement				
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_257] 5.8.0-11 Die Verantwortung für die Verfügbarkeit des vom AN gelieferten Systems liegt beim AN.	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00 DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test	
		See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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3.3.4.9 POWER SUPPLY

EL4-DIL-SSS-REQ-00308 Power supply - UPS Design (Draft)	
The three phases of each [UPS] shall be connected with the corresponding [PDU] in a way to tolerate [HW] failures if possible.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> GCR <i>Originator:</i> Product Management <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00422 Power supply - PDU Design (Draft)	
Each [PDU] shall be separately protected.	
<i>Comments:</i> •This means that an outage of one power supply phase has no impact on other phases. •This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> GCR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00309 Power supply - Availability related Components (Draft) - Redundancy			
All critical [HW] components which are related to availability (server, router, switches), shall be equipped with two independent power supplies.			
<i>Comments:</i>		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i>	GCR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Product Management	<i>Origin:</i>	
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) IPS HW DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	Integration test
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00315 Power Supply - Non-Availability related (Draft) Components - Monitoring and Backup			
Non-availability related components shall do with one power supply, but in that case monitoring and backup equipment for automatic switch-over shall be provided.			
<i>Comments:</i>		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i>	GCR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i>	
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

3.3.4.10 NETWORK AND SWITCHES

EL4-DIL-SSS-REQ-00307 Availability - Network - Overall (Draft)			
The network availability of a [DIL] location shall be at least 99.9999%.			
<i>Comments:</i>		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i>	
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Verification:</i>	Analysis
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00310 Availability - Switches - Connection with Servers (Draft)			
A server within a server cabinet shall be connected to each switch within this server cabinet.			
<i>Comments:</i>		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i>	GCR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Product Management	<i>Origin:</i>	
<i>Target Release:</i>	PV_EL_GEN_2.20.0	<i>Scope:</i>	System
	BL_OBB_40.00		
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL)	<i>Verification:</i>	Integration test
	INTEGRATED PLATFORM	<i>See also:</i>	
	SIGNALLING (IPS)		
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00311 Availability - Switches - Internal Network - (Draft) Connection with Routers			
The switches for the internal network within a cabinet shall be connected with the two routers of the network cabinet.			
<i>Comments:</i>		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i>	GCR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Product Management	<i>Origin:</i>	
<i>Target Release:</i>	PV_EL_GEN_2.20.0	<i>Scope:</i>	System
	BL_OBB_40.00		
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL)	<i>Verification:</i>	Integration test
	INTEGRATED PLATFORM	<i>See also:</i>	
	SIGNALLING (IPS)		
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00312 Availability - Switches - External Network - (Draft) Connection with Routers			
The switches for the external network within a cabinet shall be connected with the two routers in the network cabinet.			
<i>Comments:</i>		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i>	GCR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Product Management	<i>Origin:</i>	
<i>Target Release:</i>	PV_EL_GEN_2.20.0	<i>Scope:</i>	System
	BL_OBB_40.00		
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL)	<i>Verification:</i>	Integration test
	INTEGRATED PLATFORM	<i>See also:</i>	
	SIGNALLING (IPS)		
<i>Issues/Decisions:</i>			

3.3.5 SAFETY

3.3.5.1 SAFETY REQUIREMENTS [DIL]

The following relevant [DIL] functions and subfunctions or safety requirements were identified in the context of [EULYNX]:

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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Anforderungs-ID	Sicherheitsanforderung
B.2	Sichern des Fahrweges Sichern
B.3	Schutz an niveaugleichen Kreuzungen
B.4	Sichern der Abstandshaltung
B.5	Sichern der Geschwindigkeit
B.5.6	
B.5.8	
B.5.9	
B.5.12	
B.6	Schutz vor Hindernissen im Fahrweg
B.7	Bedienung und Anzeige
B.7.6	
B.8	
B.9	Überwachen des Betriebes

Table 4: DIL: Safety Requirements

== **TODO** ==

Update this picture.

Threshold values for availability [THR] were determined for each individual subsystem and for each corresponding interface based on the safety requirements of the [DIL]. In accordance with [OEVE/OENORM EN 50129], each safety-related subsystem and the interfaces were assigned a safety integrity level [SIL].

These are:

Alle Betriebsmodi inkl. Zugfahrt																											
Anforderungs-ID	Teilsysteme										Umsysteme (Ports)					interne Ports				externe Ports							
	DSTW-Zentraleinheit	Wartung und Diagnose		Lichtsignal	Gleisfreimeldung		Weiche	Ein-/Ausgabeelement	PoS		SCWS (X25oRaSTA)	RBC (X25oRaSTA)	EKSA (SCI-LX)	TCS (X25oRaSTA)	Nachbarstellwerk (X25oRaSTA/X25-Block)	CDC (EIL 2)	SCI-P	SCI-HO	SCI-LS	SCI-TDS	p3	O2	O3	LS2	TDS2	SCI-ACS	
B.2	SIL 4	-	-	SIL 3	SIL 4	-	SIL 4	SIL 4	-	SIL 4	-	SIL 4	BI	SIL 4	-	-	SIL 4	SIL 4	SIL 3	SIL 4	SIL 4	SIL 4	SIL 4	SIL 4	SIL 4	SIL 4	
B.3	SIL 4	-	-	-	-	-	-	SIL 4	SIL 4	-	-	-	-	-	SIL 4	-	-	SIL 4	-	-	-	-	SIL 4	SIL 4	-	-	-
B.4	SIL 4	-	-	SIL 4	SIL 4	-	-	SIL 4	SIL 4	-	SIL 4	-	-	SIL 4	-	-	-	-	SIL 4	SIL 4	-	-	-	SIL 4	SIL 4	SIL 4	
B.5	SIL 4	-	-	SIL 4	-	-	-	SIL 4	SIL 4	-	SIL 4	-	-	SIL 4	-	-	-	-	SIL 4	-	-	-	-	-	SIL 4	-	-
B.6	SIL 4	-	-	-	-	-	-	SIL 4	SIL 4	-	-	SIL 3	SIL 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-
B.7	SIL 4	-	-	-	-	-	-	SIL 4	SIL 4	-	-	-	SIL 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-
B.8	BI	-	-	-	-	-	-	BI	-	-	-	-	BI	-	-	-	-	-	-	-	-	-	-	-	-	-	-
B.9	BI	-	-	-	-	-	-	BI	-	-	-	-	BI	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 5: SIL-Allocation of safety-related Part Systems and Interfaces

== **TODO** ==

Update this picture.

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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3.3.5.2 TOLERABLE HAZARD RATES

EL4-DIL-SSS-REQ-00588 Safety - THR - VIL		(Draft)
The [THR] value of the subsystem [VIL] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_158] 6.2.0-1 Der THR Wert des Teilsystems VSTW muss kleiner gleich 1.0E-9 1/h sein
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Verification:</i> System test
<i>See also:</i>		
<i>Issues/Decisions:</i>		

EL4-DIL-SSS-REQ-00589 Safety - THR - LS		(Draft)
The [THR] value of the part system [LS] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_160] 6.3.0-1 Der THR Wert des Teilsystems LS muss kleiner gleich 1.0E-9 1/h sein
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-LS	<i>Verification:</i> System test
<i>See also:</i>		
<i>Issues/Decisions:</i>		

EL4-DIL-SSS-REQ-00590 Safety - THR - P		(Draft)
The [THR] value of the part system [P] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_162] 6.4.0-1 Der THR Wert des Teilsystems Weiche muss kleiner gleich 1.0E-9 1/h sein
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i> System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-P	<i>Verification:</i> System test
<i>See also:</i>		
<i>Issues/Decisions:</i>		

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00591 Safety - THR - TDS		(Draft)
The [THR] value of the part system [TDS] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
Category: Originator:	SAR Customer	Priority: Origin:
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-TDS	Verification: See also:
<i>Issues/Decisions:</i>		

EL4-DIL-SSS-REQ-00592 Safety - THR - IO		(Draft)
The [THR] value of the part system [IO] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
Category: Originator:	SAR Customer	Priority: Origin:
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) OC-IO	Verification: See also:
<i>Issues/Decisions:</i>		

EL4-DIL-SSS-REQ-00593 Safety - THR - LCSi		(Draft)
The [THR] value of the [LCSi] Converter shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
Category: Originator:	SAR Customer	Priority: Origin:
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:
Allocation:	DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE) IFC-LCSi	Verification: See also:
<i>Issues/Decisions:</i>		

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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3.3.5.3 INTERFACE - INTEGRITY LEVELS

EL4-DIL-SSS-REQ-00735 Safety - Integrity Levels - Interface - X25oRaSTA, (In negotiation) EULYNX - Functional Hazard Rates			
The use of the required interfaces ([X25oRaSTA], [EULYNX]) shall be carried out in such a way that the functional hazard rates of the involved subsystems are not increased.			
Comments: == MISMATCH == Das ist eine Anforderung, aber im LH nur als "Info" markiert.			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_170] 6.8.0-1 Der Einsatz der erforderlichen Schnittstellen (X25oRaSTA, EULYNX) muss derart erfolgen, dass...
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test
See also:			
Issues/Decisions:			

EL4-DIL-SSS-REQ-00683 Safety - Integrity Levels - Interface - Adjacent Systems (Draft)			
The interfaces of the [VIL] to the adjacent systems [ILS], [RBC], [EKSA], [SCWS] (Trackworker Safety System) and [OMC] ([TCS]) shall meet the requirements of Safety Integrity Level 4 [SIL]4.			
Comments:			
Category:	SAR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_171] 6.8.0-2 Die Schnittstellen des VSTW zu den Umsystemen ILS, RBC, EKSA, SCWS (Trackworker Safety Systeme...
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	System test
See also:			
Issues/Decisions:			

EL4-DIL-SSS-REQ-00684 Safety - Integrity Levels - Interface - OC (Draft)			
The interfaces of the [VIL] [SCI-LS], [SCI-TDS], [SCI-P], [SCI-IO] to the [OC]s light signal, track vacancy detection, points, input/output element shall fulfill the requirements according to Safety Integrity Level 4 [SIL]4.			
Comments:			
Category:	SAR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_172] 6.8.0-3 Die Schnittstellen des VSTW SCI-LS, SCI-TDS, SCI-P, SCI-IO zu den OCs Lichtsignal, Gleisfrei...
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) DIGITAL FIELD ELEMENTS (DFE)	Verification:	System test
See also:			
Issues/Decisions:			

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00685 Safety - Integrity Levels - Interface - DDSP, PS (Draft)			
The [VIL] - [DDSP] and [VIL] - [PS] interfaces shall meet the requirements for basic integrity [BI].			
<i>Comments:</i>			
<i>Category:</i>	SAR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_174] 6.9.0-1 Die Schnittstellen VSTW - DDSP und VSTW - Stromversorgung müssen Anforderungen gemäß der Bas...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00686 Safety - Integrity Levels - Development DIL (Draft)			
The development of the [DIL] shall be carried out in accordance with the safety requirement level [SIL]4 and the procedures and techniques requested in [OEVE/OENORM EN 50129] shall be taken into account.			
<i>Comments:</i>			
<i>Category:</i>	SAR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_175] 6.9.0-2 Die Entwicklung des DSTW muss gemäß der Sicherheitsanforderungsstufe SIL4 erfolgen und die d...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-DIL-SSS-REQ-00687 Safety - Integrity Levels - SW VIL, OC (Draft)			
The software of the [VIL] and the [OC]s shall be developed in accordance with the specified software safety requirements and the procedures and techniques requested in [OEVE/OENORM EN 50128] for it.			
<i>Comments:</i>			
<i>Category:</i>	SAR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_176] 6.9.0-3 Die Software-Entwicklung des VSTW und der OCs muss gemäß der festgelegten Software-Sicherhei...
<i>Target Release:</i>	PV_EL_GEN_2.20.0 BL_OBB_40.00	<i>Scope:</i>	System
<i>Allocation:</i>	DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Verification:</i>	System test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00688		Safety - Integrity Levels - Certificates		(Draft)
The supplier/manufacturer of the [DIL] shall prepare the safety certificates in accordance with [OEVE/OENORM EN 50129].				
Comments:				
Category:	SAR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_177] 6.9.0-4 Der Lieferant/Hersteller des DSTW muss die Sicherheitsnachweise gemäß [OEVE/OENORM EN 50129]...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00689		Safety - Integrity Levels - Data Communication		(Draft)
For data communication via open and closed networks, [OEVE/OENORM EN 50159] shall be complied with.				
Comments:				
Category:	SAR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_178] 6.9.0-5 Für Datenkommunikation über offene und geschlossene Netze muss [OEVE/OENORM EN 50159] eingeh...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	VIRTUAL INTERLOCKING (VIL)			
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

EL4-DIL-SSS-REQ-00690		Safety - Integrity Levels - Misuse of Track Elements		(Draft)
The supplier/manufacturer of the [DIL] shall agree and define measures with the customer to protect against misuse of the trackside elements.				
Comments:				
Category:	SAR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_179] 6.9.0-6 Der Lieferant/Hersteller des DSTW muss Maßnahmen zum Schutz vor Missbrauch der streckenseiti...	
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
Allocation:	BL_OBB_40.00	Verification:	System test	
	DIGITAL INTERLOCKING (DIL)	See also:		
	DIGITAL FIELD ELEMENTS (DFE)			
Issues/Decisions:				

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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3.3.5.4 MAINTENANCE

EL4-DIL-SSS-REQ-00691 Safety - Maintenance - SRACs (Draft)	
Any additional expenses or [SRAC]s for the maintenance personnel of the customer as a result of the [CENELEC] phases 5-10 shall be agreed between the supplier and the customer.	
<i>Comments:</i>	
Category: SAR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_181] 6.10.0-1 Für aus den CENELEC-Phasen 5-10 resultierende zusätzlichen Aufwendungen oder SRACs an das I...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL)	Verification: System test See also:
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00692 Safety - Maintenance - Activities (Draft)	
It shall not be possible for the performance of planned maintenance activities to result in unsafe conditions in terms of safety or operation.	
<i>Comments:</i>	
Category: SAR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_182] 6.10.0-2 Durch die Durchführung von geplanten Instandhaltungstätigkeiten dürfen keine sicherungstech...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL)	Verification: System test See also:
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00693 Safety - Maintenance - Exchange of FES elements (Draft)	
The exchange of systems, subsystems or components of the customer in [FES] or server racks of the supplier shall not require any safety-related training or education.	
<i>Comments:</i> The customer does not consider [DIL] (basic) training according to the supplier's concept as safety-related training or education.	
Category: SAR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 PH] [02_253] 6.10.0-3 Eine DSTW-(Basis)Schulung laut Konzept des AN ist nicht als sicherungstechnische Schulung o...
Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope: System
Allocation: DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	Verification: System test See also:
<i>Issues/Decisions:</i>	

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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3.3.5.5 MISCELLANEOUS

EL4-DIL-SSS-REQ-00694 Safety - Power Supply - Shutdown (Draft)	
In case of a power failure, the [VIL] and all components of the outdoor system shall enter a safe state.	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_184] 6.11.0-1 Bei Ausfall der Stromversorgung muss das VSTW und jegliche Komponenten der Außenanlage den ... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00411 Safety - HW Components - Electrical Safety (Draft)	
All [HW] components shall fulfill requirements for electrical safety according to [EN 62368-1].	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> SAR <i>Originator:</i> Product Management <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> Qualification test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00412 Safety - Data Configuration Process (Draft)	
The supplier shall provide a certified [CENELEC]-compliant data configuration process for the [VIL] subsystem, the [OC] subsystems and all interfaces to the sub-systems and peripheral systems.	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_188] 6.13.0-1 Der AN muss einen zertifizierten CENELEC-konformen Datenkonfigurationsprozess für das Systeme... <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00413		Safety - Safety Case		(Draft)			
The Safety Case shall have an unlimited validity.							
Comments:							
Category:		SAR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_190] 6.14.0-1 Die Gültigkeit des Sicherheitsnachweises darf zeitlich nicht beschränkt sein.	
Target Release:		PV_EL_GEN_2.20.0		Scope:		System	
Allocation:		BL_OBB_40.00		Verification:		System test	
		DIGITAL INTERLOCKING (DIL)		See also:			
		VIRTUAL INTERLOCKING (VIL)					
		INTEGRATED PLATFORM					
		SIGNALLING (IPS)					
		DIGITAL FIELD ELEMENTS (DFE)					
Issues/Decisions:							

3.3.6 MISCELLANEOUS

EL4-DIL-SSS-REQ-00654		RAMS - Diagnosis - Data for RAM Statistics		(Draft)	
The [DIL-CU] shall provide data from which RAM statistics can be created, based on the following events:					
•Starting of a [VIL] instance •Stopping of a [VIL] instance together with the reason for stopping •All state transitions of the remote control lines (separate for each line)					
Comments:		•The [DIL-CU] does not need to create any such statistics itself. •This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
Category:	MDR	Priority:	Mandatory		
Originator:	RAM	Origin:			
Target Release:	PV_EL_GEN_2.20.0	Scope:	System		
	BL_OBB_40.00				
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	Integration test		
	VIRTUAL INTERLOCKING (VIL)		Subit test		
		See also:			
Issues/Decisions:					

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00416 RAMS - VIL - Resource Allocation		(Draft)
After reintegration of a server, the automatic resource allocation of running [VIL] redundancy instances shall be avoided. Allocation shall be triggered manually at agreed timeframes to mitigate operational impact.		
Comments: •This could be e.g. after 2-3 days. •This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category: FUR Originator: RAM Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Priority: Mandatory Origin: Scope: System Verification: Integration test Subit test See also:	
Issues/Decisions:		

EL4-DIL-SSS-REQ-00417 RAMS - VIL - Automatic Restart		(Draft)
All [VIL]s affected by a hardware failure shall be restarted without manual intervention on a [CN] which is configured to run these [VIL] instance(s).		
Comments:		
Category: FUR Originator: RAM Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Priority: Mandatory Origin: Scope: System Verification: Integration test Subit test See also:	
Issues/Decisions:		

EL4-DIL-SSS-REQ-00418 RAMS - VIL - Unified Restart		(Draft)
Shutdown and reintegration of a component (e.g. [VIL]) shall be provided in a unified customer-friendly way as well as for technical analyses.		
Comments: •The data are then filtered and revised offline to make them available for the customer and to determine maintenance measures, •This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category: FUR Originator: RAM Target Release: PV_EL_GEN_2.20.0 BL_OBB_40.00 Allocation: DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Priority: Mandatory Origin: Scope: System Verification: System test See also:	
Issues/Decisions:		

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00419 RAMS - DIL - Mapping of Component Failure (Draft)	
The [DIL] shall provide a mapping of the failed hardware component to the severity level of the availability class to derive the operative consequences.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL) INTEGRATED PLATFORM SIGNALLING (IPS) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM) DIGITAL FIELD ELEMENTS (DFE)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00420 RAMS - DIL - Mapping of corrective Maintenance Action (Draft)	
The [DIL] shall provide a mapping of the corrective maintenance activities to the severity level of the availability class to derive the operative consequences.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00421 RAMS - DIL - Mapping of Severity Levels (Draft)	
The [DIL] shall provide a mapping of the severity levels to the failed / reintegrated components.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) ADMINISTRATION, DIAGNOSTIC AND MAINTENANCE-SERVICES (ADM)	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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EL4-DIL-SSS-REQ-00423		RAMS - VIL - Anti-Affinity Rule - Physical Servers		(Draft)
If possible, [VIL] redundancy instances belonging to the same region shall reside on different physical servers.				
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category:	DPR	Priority:	Mandatory	
Originator:	RAM	Origin:		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
	BL_OBB_40.00			
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test	
	VIRTUAL INTERLOCKING (VIL)	See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00424		RAMS - VIL - Equi-Distribution Rule - Physical Servers		(Draft)
The [VIL] redundancy instances shall be equally distributed to different physical servers.				
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category:	DPR	Priority:	Mandatory	
Originator:	RAM	Origin:		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
	BL_OBB_40.00			
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test	
	VIRTUAL INTERLOCKING (VIL)	See also:		
Issues/Decisions:				

EL4-DIL-SSS-REQ-00425		RAMS - VIL - Equi-Distribution Rule - Availability Zones		(Draft)
If possible, [VIL] redundancy instances belonging to the same region shall be equally distributed over the three availability zones.				
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category:	DPR	Priority:	Mandatory	
Originator:	RAM	Origin:		
Target Release:	PV_EL_GEN_2.20.0	Scope:	System	
	BL_OBB_40.00			
Allocation:	DIGITAL INTERLOCKING (DIL)	Verification:	System test	
	VIRTUAL INTERLOCKING (VIL)	See also:		
Issues/Decisions:				

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
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EL4-DIL-SSS-REQ-00426 RAMS - Sub Station - Specific Services - Affinity (Draft) Rule - Physical Servers			
The group of services which belongs to a specific sub-station shall be executed on the same physical server.			
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
Category:	DPR	Priority:	Mandatory
Originator:	RAM	Origin:	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	System test
Issues/Decisions:		See also:	

EL4-DIL-SSS-REQ-00427 RAMS - Sub Station - Specific Services - Anti- (Draft) Affinity Rule - Physical Servers			
If possible, sub-station specific service groups belonging to the same cell shall reside on different physical servers.			
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
Category:	DPR	Priority:	Mandatory
Originator:	RAM	Origin:	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) INTEGRATED PLATFORM SIGNALLING (IPS)	Verification:	System test
Issues/Decisions:		See also:	

EL4-DIL-SSS-REQ-00428 RAMS - Sub Station - Specific Services - Equi- (Draft) Distribution Rule - Physical Servers			
The sub-station specific service groups shall be equally distributed to different physical servers.			
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
Category:	DPR	Priority:	Mandatory
Originator:	RAM	Origin:	
Target Release:	PV_EL_GEN_2.20.0 BL_OBB_40.00	Scope:	System
Allocation:	DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	Verification:	System test
Issues/Decisions:		See also:	

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EL4-DIL-SSS-REQ-00429 RAMS - Sub Station - Specific Services - Equi-Distribution Rule - Availability Zones (Draft)	
If possible, sub-station specific service groups belonging to the same cell shall be equally distributed over the three availability zones.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00430 RAMS - Global Services - Equi-Distribution Rule (Draft) - Physical Servers	
The global service instances shall be equally distributed to different physical servers.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-DIL-SSS-REQ-00431 RAMS - Global Services - Equi-Distribution Rule (Draft) - Availability Zones	
If possible, global service instances shall be equally distributed over the three availability zones.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> PV_EL_GEN_2.20.0 BL_OBB_40.00 <i>Allocation:</i> DIGITAL INTERLOCKING (DIL) VIRTUAL INTERLOCKING (VIL)	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> System test <i>See also:</i>
<i>Issues/Decisions:</i>	

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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4. HISTORY

4.1 EDITION 01

4.1.1 PROPOSAL 01

Please consider the following guidelines for the review of the document:

This is the **very first version** of the system SSS and **far away from being complete!**

The current version of the SSS covers **about 20%** of the LH requirements.

The **main focus** is on the requirements on system level concerning the **component “ELEKTRA Generic System” within the subsystem VIL** (according to [LH_PART_D] D2.5 “DSTW-ZE”).

In addition this version of the SSS includes

- The **general system requirements** (according to [LH_PART_D] D2.3 “Allgemeine Anforderungen”)
- The **RAMS requirements** (according to [LH_PART_D] D2.2 “RAMS”)

The **next versions** of the SSS will then - in coordination with the analysis progress of the subsystems and the project priorities – set the focus on the following topics:

- IPSS
- MUX-System
- DFE
- ELEKTRA Application

The **verification measures** of the requirements will also be reviewed separately. But feel free to make your comments already now.

References and abbreviated terms are marked with square brackets. They will be later replaced by hyperlinks (after the transfer to Polarion™)

The review audience is not complete due to the limited scope of this first version, e. g. concerning the verification of the requirements. Nevertheless, if you think, someone is missing, please let me know.

The review **shall not result in a collection of hundreds of medium and grave comments**. If serious objections concerning the current structure and the content of the document exist, please let us clarify this better in an online discussion.

@E. Stemmer, @P. Bargiel: In your role as SEM / WMP VIL, please regard this review invitation as an information, but feel free to make your comments.

IPSS (@N. Karmatska, @C. Ruggenthaler): Please note that in a further version of this SSS the main focus will be on IPSS. Nevertheless feel free to make your comments now.

DFE (E. Brenner, M. Goldenits): Please note that in a further version of this SSS the main focus will be on DFE. Nevertheless feel free to make your comments now.

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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APPENDIX A DECISIONS

This chapter shall contain important decisions made in connection with the specified requirements.

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ANV: 3BU 17300 1251 DTAPA			

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APPENDIX B INDEX OF REQUIREMENTS

EL4-DIL-SSS-REQ-0021754	EL4-DIL-SSS-REQ-0032288	EL4-DIL-SSS-REQ-00346 100	EL4-DIL-SSS-REQ-00369116
EL4-DIL-SSS-REQ-0029983	EL4-DIL-SSS-REQ-0032389	EL4-DIL-SSS-REQ-00347 101	EL4-DIL-SSS-REQ-00374117
EL4-DIL-SSS-REQ-0030083	EL4-DIL-SSS-REQ-0032489	EL4-DIL-SSS-REQ-00348 101	EL4-DIL-SSS-REQ-00375117
EL4-DIL-SSS-REQ-0030183	EL4-DIL-SSS-REQ-0032589	EL4-DIL-SSS-REQ-00349 102	EL4-DIL-SSS-REQ-00376117
EL4-DIL-SSS-REQ-0030284	EL4-DIL-SSS-REQ-0032690	EL4-DIL-SSS-REQ-00350 102	EL4-DIL-SSS-REQ-00377118
EL4-DIL-SSS-REQ-0030384	EL4-DIL-SSS-REQ-0032790	EL4-DIL-SSS-REQ-00351 103	EL4-DIL-SSS-REQ-00406 109
EL4-DIL-SSS-REQ-0030484	EL4-DIL-SSS-REQ-0032892	EL4-DIL-SSS-REQ-00352 103	EL4-DIL-SSS-REQ-00407 109
EL4-DIL-SSS-REQ-0030585	EL4-DIL-SSS-REQ-0032993	EL4-DIL-SSS-REQ-00353 103	EL4-DIL-SSS-REQ-00408110
EL4-DIL-SSS-REQ-0030685	EL4-DIL-SSS-REQ-0033093	EL4-DIL-SSS-REQ-00354 104	EL4-DIL-SSS-REQ-00409110
EL4-DIL-SSS-REQ-00307130	EL4-DIL-SSS-REQ-0033194	EL4-DIL-SSS-REQ-00355 104	EL4-DIL-SSS-REQ-00410111
EL4-DIL-SSS-REQ-00308129	EL4-DIL-SSS-REQ-0033294	EL4-DIL-SSS-REQ-00356 104	EL4-DIL-SSS-REQ-00411139
EL4-DIL-SSS-REQ-00309130	EL4-DIL-SSS-REQ-0033395	EL4-DIL-SSS-REQ-00357 105	EL4-DIL-SSS-REQ-00412139
EL4-DIL-SSS-REQ-00310131	EL4-DIL-SSS-REQ-0033495	EL4-DIL-SSS-REQ-00358 105	EL4-DIL-SSS-REQ-00413140
EL4-DIL-SSS-REQ-00311131	EL4-DIL-SSS-REQ-0033596	EL4-DIL-SSS-REQ-00359 106	EL4-DIL-SSS-REQ-00416141
EL4-DIL-SSS-REQ-00312131	EL4-DIL-SSS-REQ-0033696	EL4-DIL-SSS-REQ-00360 106	EL4-DIL-SSS-REQ-00417141
EL4-DIL-SSS-REQ-00313123	EL4-DIL-SSS-REQ-0033796	EL4-DIL-SSS-REQ-00361 107	EL4-DIL-SSS-REQ-00418141
EL4-DIL-SSS-REQ-00315130	EL4-DIL-SSS-REQ-0033897	EL4-DIL-SSS-REQ-00362114	EL4-DIL-SSS-REQ-00419142
EL4-DIL-SSS-REQ-0031686	EL4-DIL-SSS-REQ-0033997	EL4-DIL-SSS-REQ-00363114	EL4-DIL-SSS-REQ-00420142
EL4-DIL-SSS-REQ-0031786	EL4-DIL-SSS-REQ-0034098	EL4-DIL-SSS-REQ-00364115	EL4-DIL-SSS-REQ-00421142
EL4-DIL-SSS-REQ-0031887	EL4-DIL-SSS-REQ-0034198	EL4-DIL-SSS-REQ-00365115	EL4-DIL-SSS-REQ-00422129
EL4-DIL-SSS-REQ-0031987	EL4-DIL-SSS-REQ-0034299	EL4-DIL-SSS-REQ-00366115	EL4-DIL-SSS-REQ-00423143
EL4-DIL-SSS-REQ-0032087	EL4-DIL-SSS-REQ-0034399	EL4-DIL-SSS-REQ-00367116	EL4-DIL-SSS-REQ-00424143
EL4-DIL-SSS-REQ-0032188	EL4-DIL-SSS-REQ-00345100	EL4-DIL-SSS-REQ-00368116	EL4-DIL-SSS-REQ-00425143

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
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ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00426..... 144	EL4-DIL-SSS-REQ-00469.....65	EL4-DIL-SSS-REQ-00580 120	EL4-DIL-SSS-REQ-00606 36
EL4-DIL-SSS-REQ-00427..... 144	EL4-DIL-SSS-REQ-00470.....66	EL4-DIL-SSS-REQ-00581 120	EL4-DIL-SSS-REQ-00607 36
EL4-DIL-SSS-REQ-00428..... 144	EL4-DIL-SSS-REQ-00471.....66	EL4-DIL-SSS-REQ-00582 121	EL4-DIL-SSS-REQ-00608 37
EL4-DIL-SSS-REQ-00429..... 145	EL4-DIL-SSS-REQ-00472.....67	EL4-DIL-SSS-REQ-00583 121	EL4-DIL-SSS-REQ-00610 52
EL4-DIL-SSS-REQ-00430..... 145	EL4-DIL-SSS-REQ-00473.....69	EL4-DIL-SSS-REQ-00584 122	EL4-DIL-SSS-REQ-00649 88
EL4-DIL-SSS-REQ-00431..... 145	EL4-DIL-SSS-REQ-00474.....69	EL4-DIL-SSS-REQ-00585 122	EL4-DIL-SSS-REQ-00650 91
EL4-DIL-SSS-REQ-00437..... 51	EL4-DIL-SSS-REQ-00475.....70	EL4-DIL-SSS-REQ-00586 123	EL4-DIL-SSS-REQ-00651 92
EL4-DIL-SSS-REQ-00439..... 52	EL4-DIL-SSS-REQ-00476.....70	EL4-DIL-SSS-REQ-00587 111	EL4-DIL-SSS-REQ-00654 140
EL4-DIL-SSS-REQ-00441..... 53	EL4-DIL-SSS-REQ-00477 71	EL4-DIL-SSS-REQ-00588 133	EL4-DIL-SSS-REQ-00655 73
EL4-DIL-SSS-REQ-00442..... 53	EL4-DIL-SSS-REQ-00478 71	EL4-DIL-SSS-REQ-00589 133	EL4-DIL-SSS-REQ-00656 74
EL4-DIL-SSS-REQ-00443..... 55	EL4-DIL-SSS-REQ-00481 72	EL4-DIL-SSS-REQ-00590 133	EL4-DIL-SSS-REQ-00657 75
EL4-DIL-SSS-REQ-00448..... 55	EL4-DIL-SSS-REQ-00482 72	EL4-DIL-SSS-REQ-00591 134	EL4-DIL-SSS-REQ-00658 75
EL4-DIL-SSS-REQ-00450..... 57	EL4-DIL-SSS-REQ-00566 107	EL4-DIL-SSS-REQ-00592 134	EL4-DIL-SSS-REQ-00659 77
EL4-DIL-SSS-REQ-00451..... 56	EL4-DIL-SSS-REQ-00567 108	EL4-DIL-SSS-REQ-00593 134	EL4-DIL-SSS-REQ-00660 77
EL4-DIL-SSS-REQ-00453..... 57	EL4-DIL-SSS-REQ-00568 109	EL4-DIL-SSS-REQ-00594 18	EL4-DIL-SSS-REQ-00661 78
EL4-DIL-SSS-REQ-00456..... 58	EL4-DIL-SSS-REQ-00569 111	EL4-DIL-SSS-REQ-00595 19	EL4-DIL-SSS-REQ-00662 78
EL4-DIL-SSS-REQ-00457 58	EL4-DIL-SSS-REQ-00570 112	EL4-DIL-SSS-REQ-00596 19	EL4-DIL-SSS-REQ-00663 79
EL4-DIL-SSS-REQ-00458..... 60	EL4-DIL-SSS-REQ-00571 112	EL4-DIL-SSS-REQ-00597 20	EL4-DIL-SSS-REQ-00664 80
EL4-DIL-SSS-REQ-00462..... 60	EL4-DIL-SSS-REQ-00572 112	EL4-DIL-SSS-REQ-00598 31	EL4-DIL-SSS-REQ-00665 80
EL4-DIL-SSS-REQ-00463..... 61	EL4-DIL-SSS-REQ-00573 113	EL4-DIL-SSS-REQ-00599 31	EL4-DIL-SSS-REQ-00666 80
EL4-DIL-SSS-REQ-00464..... 63	EL4-DIL-SSS-REQ-00574 114	EL4-DIL-SSS-REQ-00600 32	EL4-DIL-SSS-REQ-00667 81
EL4-DIL-SSS-REQ-00465..... 63	EL4-DIL-SSS-REQ-00575 118	EL4-DIL-SSS-REQ-00601 32	EL4-DIL-SSS-REQ-00668 81
EL4-DIL-SSS-REQ-00466..... 64	EL4-DIL-SSS-REQ-00577 119	EL4-DIL-SSS-REQ-00603 33	EL4-DIL-SSS-REQ-00669 81
EL4-DIL-SSS-REQ-00467 64	EL4-DIL-SSS-REQ-00578 119	EL4-DIL-SSS-REQ-00604 33	EL4-DIL-SSS-REQ-00670 82
EL4-DIL-SSS-REQ-00468..... 65	EL4-DIL-SSS-REQ-00579 119	EL4-DIL-SSS-REQ-00605 35	EL4-DIL-SSS-REQ-00671 124

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00672..... 124	EL4-DIL-SSS-REQ-00700.....59	EL4-DIL-SSS-REQ-00753..... 24	EL4-DIL-SSS-REQ-00778..... 40
EL4-DIL-SSS-REQ-00673..... 125	EL4-DIL-SSS-REQ-00716.....73	EL4-DIL-SSS-REQ-00754..... 25	EL4-DIL-SSS-REQ-00779..... 41
EL4-DIL-SSS-REQ-00674..... 125	EL4-DIL-SSS-REQ-00730.....76	EL4-DIL-SSS-REQ-00755..... 25	EL4-DIL-SSS-REQ-00780..... 41
EL4-DIL-SSS-REQ-00675..... 126	EL4-DIL-SSS-REQ-00731.....79	EL4-DIL-SSS-REQ-00756..... 26	EL4-DIL-SSS-REQ-00781..... 41
EL4-DIL-SSS-REQ-00676..... 126	EL4-DIL-SSS-REQ-00732.....68	EL4-DIL-SSS-REQ-00757..... 26	EL4-DIL-SSS-REQ-00782..... 42
EL4-DIL-SSS-REQ-00678..... 127	EL4-DIL-SSS-REQ-00733.....67	EL4-DIL-SSS-REQ-00758..... 26	EL4-DIL-SSS-REQ-00783..... 42
EL4-DIL-SSS-REQ-00679..... 127	EL4-DIL-SSS-REQ-00734.....68	EL4-DIL-SSS-REQ-00759..... 27	EL4-DIL-SSS-REQ-00784..... 42
EL4-DIL-SSS-REQ-00680..... 128	EL4-DIL-SSS-REQ-00735.....135	EL4-DIL-SSS-REQ-00760..... 28	EL4-DIL-SSS-REQ-00785..... 43
EL4-DIL-SSS-REQ-00681..... 128	EL4-DIL-SSS-REQ-00736.....46	EL4-DIL-SSS-REQ-00761..... 28	EL4-DIL-SSS-REQ-00786..... 43
EL4-DIL-SSS-REQ-00683..... 135	EL4-DIL-SSS-REQ-00737.....12	EL4-DIL-SSS-REQ-00762..... 28	EL4-DIL-SSS-REQ-00787..... 43
EL4-DIL-SSS-REQ-00684..... 135	EL4-DIL-SSS-REQ-00738.....15	EL4-DIL-SSS-REQ-00763..... 29	EL4-DIL-SSS-REQ-00788..... 44
EL4-DIL-SSS-REQ-00685..... 136	EL4-DIL-SSS-REQ-00739.....15	EL4-DIL-SSS-REQ-00764..... 29	EL4-DIL-SSS-REQ-00789..... 44
EL4-DIL-SSS-REQ-00686..... 136	EL4-DIL-SSS-REQ-00740.....16	EL4-DIL-SSS-REQ-00765..... 29	EL4-DIL-SSS-REQ-00790..... 44
EL4-DIL-SSS-REQ-00687..... 136	EL4-DIL-SSS-REQ-00741.....16	EL4-DIL-SSS-REQ-00766..... 30	EL4-DIL-SSS-REQ-00793..... 47
EL4-DIL-SSS-REQ-00688..... 137	EL4-DIL-SSS-REQ-00742.....17	EL4-DIL-SSS-REQ-00767..... 30	EL4-DIL-SSS-REQ-00794..... 47
EL4-DIL-SSS-REQ-00689..... 137	EL4-DIL-SSS-REQ-00743.....17	EL4-DIL-SSS-REQ-00768..... 30	EL4-DIL-SSS-REQ-00795..... 47
EL4-DIL-SSS-REQ-00690..... 137	EL4-DIL-SSS-REQ-00744.....18	EL4-DIL-SSS-REQ-00769..... 34	EL4-DIL-SSS-REQ-00796..... 48
EL4-DIL-SSS-REQ-00691..... 138	EL4-DIL-SSS-REQ-00745.....20	EL4-DIL-SSS-REQ-00770..... 34	EL4-DIL-SSS-REQ-00797..... 48
EL4-DIL-SSS-REQ-00692..... 138	EL4-DIL-SSS-REQ-00746.....20	EL4-DIL-SSS-REQ-00771..... 34	EL4-DIL-SSS-REQ-00798..... 48
EL4-DIL-SSS-REQ-00693..... 138	EL4-DIL-SSS-REQ-00747.....21	EL4-DIL-SSS-REQ-00772..... 35	EL4-DIL-SSS-REQ-00799..... 49
EL4-DIL-SSS-REQ-00694..... 139	EL4-DIL-SSS-REQ-00748.....21	EL4-DIL-SSS-REQ-00773..... 37	EL4-DIL-SSS-REQ-00800..... 49
EL4-DIL-SSS-REQ-00695..... 61	EL4-DIL-SSS-REQ-00749.....22	EL4-DIL-SSS-REQ-00774..... 38	EL4-DIL-SSS-REQ-00801..... 50
EL4-DIL-SSS-REQ-00696..... 62	EL4-DIL-SSS-REQ-00750.....22	EL4-DIL-SSS-REQ-00775..... 38	EL4-DIL-SSS-REQ-00802..... 45
EL4-DIL-SSS-REQ-00697..... 62	EL4-DIL-SSS-REQ-00751.....23	EL4-DIL-SSS-REQ-00776..... 40	EL4-DIL-SSS-REQ-00803..... 45
EL4-DIL-SSS-REQ-00699.....59	EL4-DIL-SSS-REQ-00752.....23	EL4-DIL-SSS-REQ-00777..... 40	EL4-DIL-SSS-REQ-00804..... 46

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			

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EL4-DIL-SSS-REQ-00805.....49	EL4-DIL-SSS-REQ-0080914	EL4-DIL-SSS-REQ-00815 39	EL4-DIL-SSS-REQ-00819 12
EL4-DIL-SSS-REQ-00806.....14	EL4-DIL-SSS-REQ-0081024	EL4-DIL-SSS-REQ-00816 39	
EL4-DIL-SSS-REQ-0080713	EL4-DIL-SSS-REQ-0081127	EL4-DIL-SSS-REQ-00817 102	
EL4-DIL-SSS-REQ-00808.....13	EL4-DIL-SSS-REQ-0081439	EL4-DIL-SSS-REQ-00818113	

END OF DOCUMENT

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~9950	01 PD 01
ANV: 3BU 17300 1251 DTAPA			